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A Comparative Study of Fraud Detection Models under Class Imbalance and Data Constraints

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Dear Readers,

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Nouha Laataoui

Rennes, June 2026

Confidentiality Declaration

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This Graduating Project does not contain confidential organisational data. All datasets used are publicly available benchmark datasets released for academic research purposes.

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Abstract

Financial fraud detection presents a fundamental machine learning challenge where fraudulent events are rare, costly, and represent only a small fraction of observed transactions. This study conducts a comparative evaluation of four machine learning models—Logistic Regression, Random Forest, XGBoost, and LightGBM—across three structurally different fraud detection datasets: the Kaggle Credit Card Fraud dataset (0.17% fraud rate), the IEEE-CIS e-commerce fraud dataset (3.50%), and the Elliptic Bitcoin transaction dataset (9.76%).

A leakage-safe experimental framework was used to avoid data contamination during preprocessing, while model performance was assessed using discrimination, calibration, robustness, and computational efficiency metrics. The study also compares multiple class imbalance handling strategies and evaluates the impact of temporal distribution shifts through a forward-chaining validation protocol.

Results show that gradient boosting models consistently outperform classical baselines across all three datasets, with LightGBM providing the strongest overall balance of predictive performance, calibration quality, robustness, and computational efficiency. In the IEEE-CIS dataset, class weighting combined with post-hoc isotonic calibration achieved both high discrimination (PR-AUC = 0.8171) and excellent probability reliability (ECE = 0.0005). Temporal evaluation in the Elliptic dataset revealed a 0.117 PR-AUC degradation under concept drift, highlighting limitations of conventional random-split evaluations.

The results show that fraud prevalence alone does not explain performance differences. Instead, dataset complexity, feature richness, missingness patterns, and temporal structure emerge as the primary drivers of model behaviour. The study provides practical guidance for selecting fraud detection models under varying operational constraints and contributes a reproducible evaluation framework for future fraud detection research.

Chapter 1: Introduction

1.1 Background and Motivation

Financial fraud is an ongoing and evolving challenge that banks, payment providers, fintech firms, e-commerce platforms, and blockchain-based financial services face. Direct monetary losses, operational investigation costs, reputation risk, customer dissatisfaction, and regulatory exposure are all risks that come with fraudulent transactions. The challenge is also dynamic: fraud trends shift as criminals adjust to countermeasures implemented, necessitating detection systems that are accurate, robust, and operationally feasible in changing environments.

From a machine learning perspective, fraud detection differs from standard classification tasks because fraudulent transactions are rare events. In many real-world datasets, fraud represents far below one percent of all observations. This creates a fundamental methodological problem: standard evaluation metrics such as accuracy can be deeply misleading. In a dataset with a fraud rate of 0.17%, a naive classifier that labels every transaction as legitimate would exceed 99% accuracy while detecting no fraud at all. Fraud detection therefore demands evaluation methods that focus directly on the minority class and on the trade-off between correctly identifying fraud and controlling the volume of false alerts.

Machine learning models are now central to fraud detection systems. Despite the success of the latest models, classical approaches such as Logistic Regression and Random Forest remain attractive due to their interpretability, relative stability, and easier deployment in regulated environments. Gradient boosting frameworks such as XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017) are generally better suited to modelling complex tabular data with non-linear interactions and heterogeneous feature effects and can produce better predictive performance. However, these models also require more careful tuning, more computational resources, and stronger governance justification. A decision-oriented comparison must therefore examine not only which model obtains the highest score on one

dataset, but how performance changes across datasets and under different operational conditions.

1.2 Research Problem and Research Questions

Existing fraud detection studies often evaluate one dataset, one imbalance strategy, and a limited set of metrics, providing limited guidance for practitioners who must operate under different fraud prevalences, data volumes, feature structures, missingness patterns, and computational constraints. This Graduating Project addresses this gap through a controlled comparative study across three fraud detection contexts spanning banking card transactions, e-commerce transactions, and cryptocurrency blockchain transactions. The study is guided by two research questions:

RQ1: How do recent fraud detection models, particularly gradient boosting frameworks, perform compared to classical baseline models on benchmark fraud datasets under class imbalance?

RQ2: What data characteristics, especially class imbalance severity, dataset size, feature complexity, missingness, and temporal structure, affect fraud detection model performance?

RQ1 is operationalised across four dimensions: discrimination performance (PR-AUC and AP), probability calibration quality (ECE), training stability across N=10 repeated experiments, and computational efficiency (training time). Evaluating all four is necessary because a model with marginal PR-AUC superiority but poor calibration and high computational cost may be worse for deployment than a marginally weaker but more reliable and faster alternative. RQ2 is operationalised through three complementary analyses: a cross-dataset comparison varying fraud rate, feature count, missingness, and domain simultaneously; imbalance sensitivity analyses that vary the test-set fraud rate while holding the model fixed; and dataset size sensitivity analyses that vary the training fraction to estimate how each model family exploits additional labelled data.

The three datasets were selected to address RQ2 by varying the five hypothesised characteristics systematically. Phase 1 (Kaggle) provides extreme imbalance (577:1) with 30 simple features and no missing values. Phase 2 (IEEE-CIS) provides moderate imbalance (27:1) with 627 heterogeneous features and 75.6% structural missingness in identity features. Phase 3 (Elliptic) provides lower imbalance (9:1) with 165 structured features, no missingness, and an explicit temporal dimension. This variation enables the study to separate the effects of fraud prevalence, feature complexity, missingness, and temporal structure — factors that co-vary in any single dataset but are varied independently across the three phases.

1.3 Research Objectives

The study pursues four interconnected objectives. First, to compare Logistic Regression, Random Forest, XGBoost, and LightGBM across three benchmark fraud datasets using PR-AUC, calibration metrics, and operational timing as primary evaluation dimensions — moving beyond the accuracy-focused comparisons that characterise much of the existing literature. Second, to determine what dataset features are contributing to performance differences between phases, specifically testing whether fraud prevalence, feature richness, missingness, training data size, and temporal structure each make independent contributions to observed performance variation. Third, to assess model robustness and stability through repeated experiments across ten random seeds, paired statistical tests with Bonferroni correction, and bootstrap confidence intervals, ensuring that reported performance differences reflect genuine model properties rather than favourable random splits. Fourth, to develop practical guidance for model and pipeline selection in fraud detection contexts where data constraints, class imbalance, computational feasibility, and probability reliability are all operational concerns; translating empirical findings into stakeholder-specific recommendations for banking, e-commerce, cryptocurrency, and regulated-environment contexts.

These objectives are designed to work in tandem rather than sequentially. The model comparison (Objective 1) only yields trustworthy conclusions if robustness is assessed (Objective 3); the dataset characteristic analysis (Objective 2) only becomes actionable if it is translated into deployment guidance (Objective 4). This interdependence is reflected in the study design: each phase is evaluated using the same protocol, and the practical recommendations in Chapter 5 are derived directly from the quantitative evidence in Chapter 4 rather than stated independently. The objectives also reflect a deliberate departure from the dominant paradigm in fraud detection benchmarking, which typically optimises a single model on a single dataset and reports accuracy. By explicitly framing objectives around calibration, robustness, and operational timing alongside discrimination, this study defines success in terms that are meaningful for practitioners deploying systems in production environments — not only for researchers seeking to maximise a benchmark score.

1.4 Contributions of this Study

This study evolved beyond a straightforward model comparison into a broader investigation of experimental validity, imbalance handling, probability calibration, and operational scalability in fraud detection. Five concrete contributions are made.

Leakage-safe preprocessing framework. A widespread but underreported flaw in fraud detection benchmarking is that preprocessing statistics — such as scaling parameters, encoding mappings, or imputation values — are often computed on the full dataset before the train-test split, allowing test-set information to contaminate the training procedure and inflate reported performance. This study addresses this by fitting all preprocessing statistics exclusively on training data and applying them as fixed, frozen transformations to held-out sets. Every reported test-set metric is therefore a genuine out-of-sample estimate, making the results directly comparable to production deployment conditions rather than optimistic laboratory benchmarks.

Reproducible and scalable pipeline. Reproducibility in machine learning research is undermined when computational cost makes re-running experiments impractical. This study introduces two caching optimisations using standard Python tools — joblib preprocessing caching and pickle-based dataset serialisation — that reduce repeated preprocessing time from 170.9 seconds to 0.09 seconds (a 1,899× acceleration) and data loading from 31.3 seconds to 2.1 seconds (15× faster). These optimisations require no specialised hardware, making the full experimental protocol reproducible by any researcher with a standard computing environment.

Controlled imbalance strategy comparison. The choice of how to handle class imbalance — whether to reweight the loss function, undersample the majority class, generate synthetic minority examples, or use balanced ensemble methods — is one of the most contested decisions in fraud detection, yet most studies test only one strategy per dataset, making it impossible to isolate the effect of the strategy itself. This study evaluates four distinct imbalance handling pipelines on the same dataset, test set, and evaluation protocol across ten repeated experiments with Bonferroni-corrected statistical tests, providing the first controlled evidence that class weighting outperforms hybrid RUS+BorderlineSMOTE oversampling on the IEEE-CIS e-commerce dataset (Wilcoxon $p = 0.0020$).

Calibration-focused evaluation. Fraud detection systems in practice do not simply classify transactions as fraudulent or legitimate — they rank alerts by predicted probability so that investigators can prioritise their limited capacity. A model whose probability outputs are poorly calibrated will mislead this ranking process regardless of how high its discrimination score appears. This study incorporates Expected Calibration Error (ECE) as a first-class evaluation metric alongside PR-AUC, and demonstrates that post-hoc isotonic calibration applied to a class weighting pipeline simultaneously achieves the highest discrimination (PR-AUC = 0.8171) and near-perfect probability reliability (ECE = 0.0005) — resolving a

discrimination–calibration trade-off that previous studies treat as inherent rather than addressable.

Cross-domain and temporal validation. Most fraud detection benchmarks evaluate a single model on a single dataset, making it unclear whether reported performance reflects genuine model properties or the specific characteristics of one dataset. This study evaluates four model families across three structurally distinct fraud detection domains — banking card transactions, e-commerce, and cryptocurrency blockchain — and includes a temporal forward-chaining evaluation in Phase 3 that simulates the operational condition of deploying a model trained on historical data to future transactions. This temporal evaluation reveals a 0.117 PR-AUC degradation under concept drift, a deployment fragility that is completely invisible under the standard random-split protocols used in most benchmark studies.

1.5 Structure of the Report

Chapter 2 reviews the literature across six thematic sections: why fraud detection is structurally difficult, why evaluation methodology choices matter, why classical machine learning models remain relevant despite their limitations, why gradient boosting frameworks have come to dominate competitive tabular classification, why imbalance handling remains an unresolved methodological question, and what specific research gaps this study addresses. Chapter 3 presents the research methodology in full, covering the three-phase experimental design, dataset selection rationale, model configurations with explicit hyperparameter justification, the leakage-safe preprocessing protocol, evaluation metrics with selection rationale, sensitivity analysis designs, and ethical considerations. Chapter 4 presents and discusses the experimental results across all three phases, with cross-phase synthesis and explicit answers to RQ1 and RQ2 in Section 4.5. Chapter 5 develops practical implications for five distinct stakeholder groups, presents the integrated deployment framework with recommendation matrix, and provides the concluding synthesis. Chapter 6 discusses six

limitations of the study with their specific implications for result interpretation and proposes seven substantiated future research directions.

Chapter 2: Literature Review

2.1 Why Fraud Detection Remains Difficult

Fraud detection has evolved from rule-based expert systems toward machine learning over the past two decades. Despite this progress, the field continues to face a set of persistent and interrelated challenges that make it structurally harder than standard supervised classification problems.

The most fundamental challenge is extreme class imbalance. Fraudulent transactions typically constitute a tiny fraction of all observations — in many banking and payment datasets, fewer than one percent. This creates a mismatch between the objectives of standard supervised learning, which minimise global loss and favour the majority class, and the operational priority of detecting the rare but costly minority class. Dal Pozzolo et al. (2015) demonstrated that this imbalance affects not only training dynamics but also probability calibration: models trained under extreme imbalance tend to produce systematically underestimated fraud probabilities, corrupting alert ranking systems that depend on reliable probability scores. This insight — that imbalance is a calibration problem as much as a classification problem — is foundational for the present study.

Beyond imbalance, real-world fraud datasets are affected by high noise and evolving patterns. Fraudsters continuously adapt their strategies in response to deployed detection mechanisms, creating what the literature calls concept drift (Rao et al., 2025). Privacy constraints compound these difficulties further, as financial institutions are typically required to anonymise features and restrict access to transactional metadata, limiting both feature engineering possibilities and the interpretability of deployed models (Luo et al., 2024).

A growing tension also exists between methodological sophistication and operational feasibility. Recent academic research increasingly explores graph neural networks (GNNs) that can theoretically capture relational dependencies missed by tabular models (Xu et al.,

2024; Zhang et al., 2025). However, privacy regulations often restrict access to the relational information GNNs require, and their computational complexity raises latency and scalability concerns incompatible with real-time transaction scoring (Li and Yang, 2023). As Rao et al. (2025) document from a large-scale production perspective, the practical adoption of GNN-based approaches remains limited, which explains why tabular gradient boosting models continue to dominate production fraud detection systems.

However, much of the evidence supporting GNN superiority originates from research benchmarks rather than large-scale operational deployments, leaving uncertainty regarding the extent to which reported gains translate into production environments with strict latency, governance, and privacy constraints.

A further challenge that receives limited attention in the academic literature is the quality and completeness of ground-truth fraud labels. In operational fraud detection, a transaction is labelled as fraudulent only after it has been investigated by a human analyst and the fraud confirmed. This investigation process introduces three structural biases. First, only transactions that were flagged by existing detection systems are investigated, meaning that fraud patterns not detected by current rules are never labelled — a form of selection bias known as feedback loop contamination. Second, investigation capacity is limited, so not all suspicious transactions are reviewed, creating under-labelling of the fraud class. Third, investigation outcomes are uncertain, and some labelled frauds are later reversed (chargebacks that turn out to be legitimate disputes). These label quality issues mean that the true fraud rate in any deployed dataset is both uncertain and likely an underestimate of the actual rate. All three public benchmark datasets used in this study have been subject to varying degrees of these processes, which is an acknowledged limitation discussed further in Chapter 6.

2.2 Why Evaluation Methodology Matters

A recurring theme in the fraud detection literature is that evaluation choices profoundly influence which models appear to perform best, and that many published comparisons are methodologically flawed in ways that limit the generalisability of their conclusions.

Metric selection is the most studied issue. Accuracy is universally recognised as inappropriate under extreme imbalance. ROC-AUC has a known weakness: it tends to remain optimistic when the minority class is rare because it aggregates performance across all thresholds independently of the positive-class proportion (Sowmiya et al., 2024; Shang et al., 2023). Precision-Recall Area Under the Curve (PR-AUC) is more appropriate because it focuses entirely on positive-class performance. Chen et al. (2025) demonstrate that PR-AUC and ROC-AUC rankings can diverge substantially on the same dataset, underscoring that metric choice is not cosmetic.

Threshold selection is a second critical but often overlooked issue. Models evaluated at the default 0.5 probability threshold — rarely optimal under class imbalance — produce F1 scores that conflate threshold choice effects with model architecture effects (de la Cruz Huayanay et al., 2024). A third critical issue is data leakage in preprocessing. When preprocessing statistics are computed on the full dataset before splitting, test-set information contaminates the training procedure, systematically inflating reported performance. This problem is particularly acute when SMOTE is applied before splitting, since synthetic samples are generated from test-set statistics the model should never have seen (Dal Pozzolo et al., 2015). Designing a genuinely leakage-free pipeline is therefore a methodological contribution in itself.

Finally, probability calibration has received limited attention in fraud detection despite its direct operational relevance. When fraud investigators prioritise alerts by predicted probability, the ranking system relies on probabilities that accurately reflect the true

likelihood of fraud. A model that assigns a fraud probability of 0.9 to a transaction that is actually legitimate 40% of the time is not merely imprecise — it is actively misleading the investigation team about where to allocate their effort. Expected Calibration Error (ECE), the metric used in this study, measures the mean absolute difference between predicted probabilities and observed frequencies across probability bins, providing a direct assessment of this operational reliability. Carcillo et al. (2023) identify calibration quality as a critical dimension that is routinely overlooked in benchmark studies, and the Phase 2 results of this study demonstrate that the calibration-discrimination trade-off can be resolved — rather than accepted as inherent — through post-hoc isotonic calibration.

Nevertheless, calibration remains inconsistently reported across fraud detection studies, making it difficult to determine whether observed performance differences reflect genuine discrimination improvements or simply differences in probability reliability.

A fifth evaluation issue — reproducibility — has received growing attention as a systemic problem in applied machine learning research. Published fraud detection results are difficult to reproduce because preprocessing pipelines, train-test splits, threshold selection procedures, and random seeds are rarely fully specified. The consequence is that two studies reporting results on the same dataset are often measuring different quantities. This study addresses reproducibility by fully specifying all preprocessing steps, split ratios, threshold selection logic, bootstrap parameters, and statistical test choices, and by implementing a caching architecture that makes exact computational reproduction feasible. The 1,899× preprocessing acceleration documented in Section 3.6 makes it practical for researchers to re-run the full experimental protocol without prohibitive computational cost.

2.3 Classical Machine Learning Models in Fraud Detection

The continued relevance of classical machine learning models in fraud detection is not primarily a question of predictive performance — gradient boosting frameworks consistently

outperform them on benchmark evaluations — but of operational and regulatory feasibility. The General Data Protection Regulation (GDPR) in the European Union (EU) grants individuals the right not to be subject to solely automated decisions with significant effects, and requires organisations to provide meaningful explanations when those decisions are challenged. This regulatory constraint transforms model interpretability from a secondary preference into a genuine technical requirement: a model whose fraud score cannot be decomposed into attributable feature contributions may be legally indefensible in production, regardless of how high its PR-AUC is. Understanding why classical models remain in use therefore requires understanding the specific regulatory and operational conditions under which their interpretability advantages outweigh their performance limitations.

Logistic Regression represents the most interpretable end of this spectrum. Its output is a probability estimate produced by applying the logistic function to a weighted sum of input features, where each weight can be directly inspected and each fraud score decomposed into individual feature contributions at inference time. This transparency satisfies regulatory attribution requirements in a way that more complex models cannot match without post-hoc approximation tools. However, interpretability comes at a fundamental cost: a linear decision boundary cannot represent interactions between features, threshold effects, or the localised patterns in feature space that characterise many fraud typologies. Prior benchmarking studies — including Singh et al. (2025) and Sowmiya et al. (2025) — consistently find Logistic Regression ranking last or near-last on imbalanced fraud datasets, a pattern replicated across all three phases of the present study. The practical implication is that Logistic Regression is best understood as a regulatory compliance baseline rather than a competitive fraud detection architecture: it defines the minimum acceptable performance floor under interpretability constraints, not the performance ceiling under operational ones.

Random Forest occupies a more nuanced position. By training an ensemble of decision trees on bootstrap samples of the training data — each using a random subset of features at each split, following Breiman (2001) — it overcomes the linear boundary limitation while retaining a degree of interpretability through mean decrease in impurity feature importance scores. These aggregate importance measures can satisfy model-level regulatory documentation requirements even when individual tree structures are too complex to inspect directly. The ensemble averaging mechanism also provides an important operational benefit in fraud detection specifically: because fraudulent transactions are rare, the absolute count of minority-class examples in any training set is small, and individual tree predictions are highly sensitive to which fraud cases happen to appear in a given bootstrap sample. Averaging across many trees substantially reduces this variance. However, Random Forest carries a significant training cost relative to gradient boosting alternatives — a gap that widens as dataset size and feature dimensionality increase — which constrains its viability in production environments requiring frequent model retraining in response to evolving fraud patterns.

The critical unresolved tension that motivates the model comparison in this study is therefore not whether classical models are useful in principle, but under what data conditions their performance deficit relative to gradient boosting is large enough to justify the additional governance burden of deploying a less interpretable model. A study that evaluates models on a single, low-dimensional dataset cannot answer this question. Determining where the performance crossover occurs — and whether it is consistent across domains — requires the kind of controlled cross-complexity comparison that Chapters 3 and 4 of this study provide.

2.4 Gradient Boosting Frameworks: XGBoost and LightGBM

Gradient boosting decision tree frameworks — principally XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017) — have become the dominant paradigm for competitive tabular classification, and their adoption in fraud detection reflects both their

empirical performance and specific architectural properties that align well with the structural challenges of imbalanced rare-event prediction.

Both frameworks build an ensemble of decision trees sequentially, where each new tree is fitted to the residual errors of the current ensemble. XGBoost advanced over classical gradient boosting by introducing column subsampling at the tree and node level, L1 and L2 regularisation terms directly in the tree-splitting objective, and efficient handling of sparse inputs and missing values — properties that are particularly valuable when fraud datasets contain high-dimensional feature spaces with structural missingness, as in the IEEE-CIS context evaluated in Phase 2.

LightGBM introduced two algorithmic innovations with direct relevance to fraud detection at scale. First, gradient-based one-side sampling (GOSS) retains all high-gradient training examples — those hardest to classify, carrying the most information about the decision boundary — while randomly subsampling the low-gradient majority. In an imbalanced fraud dataset, high-gradient examples are disproportionately fraudulent transactions and borderline legitimate transactions near the decision boundary: exactly the cases where the model needs the most learning signal. Second, leaf-wise (best-first) tree growth selects the single leaf that would maximally reduce training loss and expands only that leaf, rather than expanding all leaves at a given depth symmetrically. In a fraud dataset where illicit patterns are localised — for example, concentrated in a specific combination of card type, merchant category, and transaction timing — leaf-wise growth rapidly specialises the tree in that region, while XGBoost's level-wise growth allocates splits across the entire feature space, including regions that carry little fraud signal.

These architectural differences generate a testable prediction: the performance gap between LightGBM and XGBoost should widen as feature space complexity increases, because the leaf-wise advantage is most valuable when discriminative patterns are localised within a high-

dimensional space. The existing literature does not resolve whether this prediction holds in practice. Al-Daoud and Abu-AlSondos (2025) report consistent gradient boosting superiority across multiple fraud benchmarks without distinguishing between the two frameworks. Singh et al. (2025) and Sowmiya et al. (2025) find that the advantage of one framework over another is dataset-dependent, but neither study systematically varies feature complexity while holding other dataset properties constant. The result is a literature in which two plausible conclusions coexist without resolution: that LightGBM is generally superior, or that the difference is negligible and context-specific. This contradiction cannot be resolved by adding another single-dataset study to the literature. It requires a controlled comparison that varies feature space complexity across datasets while keeping the evaluation protocol constant — which is precisely the design adopted in Phases 1, 2, and 3 of this study. The empirical results confirm the architectural prediction: the XGBoost–LightGBM PR-AUC gap grows from a statistically negligible 0.004 on the low-dimensional Kaggle dataset (30 features, $p = 0.850$) to a substantial and significant 0.085 on the high-dimensional IEEE-CIS dataset (627 features), providing the first controlled cross-complexity confirmation of this conditionality.

Both frameworks also include native class imbalance mechanisms. XGBoost's scale_pos_weight multiplies the gradient contribution of positive-class examples by a scalar proportional to the inverse class ratio. LightGBM's class_weight='balanced' applies equivalent balanced weighting. Critically, both mechanisms adjust the learning dynamics without altering the training data distribution, so the resulting probability estimates remain anchored to the true class frequency — unlike SMOTE-based methods, which shift the empirical class proportion the model observes during training and thereby damage probability calibration at inference time. This calibration advantage of class weighting over resampling, first identified theoretically by Dal Pozzolo et al. (2015) and confirmed empirically in the Phase 2 results of this study, is examined in detail in Section 2.5.

2.5 Imbalance Handling: An Unresolved Research Question

The optimal strategy for handling class imbalance in fraud detection remains actively debated, partly because the theoretical properties of each strategy interact with model architecture, dataset characteristics, and evaluation metrics in complex ways that single-strategy studies cannot disentangle.

Class weighting modifies the loss function by multiplying the gradient contribution of minority-class examples by a factor inversely proportional to their frequency. Critically, the training data distribution itself is unchanged — the model still observes 577 legitimate transactions for every fraudulent one in the Phase 1 context — which means that the empirical class proportions that anchor probability calibration remain intact. This is the mechanistic reason class weighting produces better-calibrated probabilities than resampling-based alternatives: probability estimates reflect the true deployment frequency of fraud rather than an artificially balanced distribution. Class weighting also preserves all majority-class information, which matters when legitimate transaction patterns provide context for identifying borderline fraud.

SMOTE and its variants — including BorderlineSMOTE, which focuses synthetic generation on minority examples near the class boundary — generate new minority examples by interpolating between existing ones in feature space. The core calibration damage mechanism identified by Dal Pozzolo et al. (2015) is as follows: when a model trains on SMOTE-augmented data, its implicit prior on class frequency reflects the augmented distribution (20% fraud in Pipeline C) rather than the deployment distribution (3.5% fraud), causing it to systematically overestimate fraud probability at inference time. On high-dimensional heterogeneous data like the IEEE-CIS dataset with 627 features including categorical variables, binary missingness indicators, and continuous amounts, linear interpolation between two fraud examples is particularly likely to produce feature combinations that are

implausible or internally inconsistent, further degrading model quality. These theoretical predictions are confirmed in the Phase 2 results: Pipeline C achieves lower PR-AUC than Pipeline A in 9 of 10 repeated experiments (Wilcoxon $p = 0.0020$).

The balanced ensemble approach, implemented through LightGBM's is_unbalance flag and Balanced Random Forest Classifier, differs from both class weighting and data-level resampling. Rather than modifying a single model's loss function or augmenting training data, it trains multiple base learners on differently resampled subsets and combines their predictions. The Phase 2 cost-sensitive evaluation confirms that Pipeline D achieves the lowest false-negative count under all tested FN: FP cost ratios, suggesting it is most appropriate when minimising missed fraud is the overriding operational objective regardless of calibration quality.

The comparative literature does not provide actionable guidance because most studies evaluate a single strategy per dataset, confounding imbalance strategy effects with dataset effects. The controlled N=10 pipeline comparison in Phase 2 of this study directly addresses this gap by isolating the imbalance strategy effect on the same dataset, test set, and evaluation protocol.

2.6 Research Gap and Positioning of this Study

Three persistent gaps motivate this study. First, most comparative studies evaluate models on a single dataset under a single imbalance strategy, making it impossible to disentangle model architecture effects from preprocessing effects or dataset-specific characteristics. Second, the relative merits of imbalance handling strategies are contested, but contradictions are rarely resolved because studies do not control for dataset and evaluation protocol effects. Third, operational dimensions — calibration quality, training efficiency, temporal robustness, and reproducibility — are underreported despite their direct relevance to deployment decisions.

This study positions itself as an operationally grounded benchmark that addresses all three gaps simultaneously. It evaluates four model families across three datasets, compares four imbalance strategies on the same controlled protocol, and systematically reports calibration, timing, temporal robustness, and statistical significance alongside discrimination metrics. Crucially, the study also documents a scalable, leakage-free preprocessing framework as a methodological contribution in its own right — addressing the gap between academic benchmark practices and the rigour needed for deployment-grade evaluation.

A fourth gap — not typically framed as such but evident from reviewing the literature — is the disconnection between evaluation protocols and operational reality. Studies that use random splits, fixed thresholds, and accuracy-based metrics are implicitly assuming that the deployment environment is stationary, that investigation capacity is unlimited, and that model probability outputs need not be reliable. In practice, all three assumptions are routinely violated. By incorporating temporal forward-chaining evaluation, max-F1 threshold selection on a held-out validation set, and Expected Calibration Error as a first-class metric, this study builds operational reality into the evaluation protocol rather than treating it as an afterthought. This design philosophy — that evaluation choices should reflect deployment conditions — is the thread that connects all five methodological contributions listed in Section 1.4.

Taken together, these four gaps point to a systemic issue in the fraud detection literature: the incentive structure of academic benchmarking rewards maximising a single metric on a widely used dataset, while the needs of practitioners require understanding how model behaviour changes across data environments and operational conditions. This divergence between what is studied and what is needed is not unique to fraud detection — it has been documented in applied machine learning more broadly — but it is particularly consequential here because deployed fraud detection systems operate under regulatory scrutiny, resource constraints, and continuous adversarial pressure that laboratory benchmarks do not capture.

By designing this study around operational validity rather than benchmark optimisation, the aim is not to replace the existing literature but to complement it with evidence that is directly interpretable in deployment terms. The findings of this study are therefore most appropriately read not as a claim that any particular model is universally superior, but as a framework for understanding under what conditions each model family is likely to be the right choice — a question that cannot be answered by any single-dataset, single-metric evaluation, however carefully executed.

Chapter 3: Research Methodology

3.1 Research Design

This study adopts a quantitative experimental research design using secondary data from publicly available fraud detection benchmarks. The quantitative approach is appropriate because the research questions concern measurable performance differences requiring numerical comparison across controlled experimental conditions. Primary data collection is not feasible in the fraud detection domain for individual researchers due to commercial confidentiality requirements and data sensitivity.

The three-phase progressive structure systematically varies the conditions under which models are compared. Phase 1 isolates the effect of extreme class imbalance on a simple, clean dataset. Phase 2 introduces the full complexity of a production e-commerce fraud context and enables a controlled imbalance strategy comparison. Phase 3 introduces temporal structure and a different domain to test the generalisability of earlier findings. Holding the evaluation protocol constant across all three phases separates dataset complexity effects from model architecture effects, enabling the study's central contribution: demonstrating that dataset complexity, not fraud prevalence, is the dominant driver of detection performance differences.

3.2 Datasets

Table 1: Overview of the three datasets used in the study.

Phase	Dataset	Domain	Transactions	Fraud Rate	Feature Structure
1	Kaggle Credit Card Fraud	Banking/card	283,726[a]	0.17%	30 (Time, Amount, V1–V28 PCA)
2	IEEE-CIS Fraud Detection	E-commerce	590,540	3.50%	627 after engineering & missingness indicators

3	Elliptic Bitcoin Transactions	Cryptocurrency	46,564 labelled	9.76%	165 anonymised (94 local + 71 aggregated)

[a] Kaggle: after removal of 1,081 exact duplicate rows from the original 284,807 transactions.

3.2.1 Phase 1 — Kaggle Credit Card Fraud Detection Dataset

The Kaggle Credit Card Fraud dataset contains 284,807 European credit card transactions with 492 labelled as fraudulent (0.1727% fraud rate; 577:1 imbalance ratio). To eliminate the exact duplicate rows from the data set which would be present in both training set and test set, 1,081 exact duplicate rows have been removed leaving 283,726 rows. Features V1–V28 are PCA-transformed by the data provider under GDPR, preventing domain-based feature engineering and limiting importance findings to relative rankings of opaque components. Amount is log-scaled during preprocessing; Time is retained as an exploratory variable only.

3.2.2 Phase 2 — IEEE-CIS Fraud Detection Dataset

The IEEE-CIS Fraud Detection dataset contains 590,540 e-commerce transactions after merging transaction and identity tables, with a 3.499% fraud rate. Identity features are available for only 24.42% of transactions, introducing structural missingness. All preprocessing statistics — missingness indicators, frequency encodings, scaling and encoding parameters — were fitted exclusively on the training partition and applied as fixed transformations to held-out sets, yielding a leakage-safe 627-feature matrix. Four imbalance handling pipelines were evaluated: Pipeline A (class weighting, full 413,378 training rows), Pipeline B (RandomUnderSampler 1:1, 28,928 rows), Pipeline C (RUS followed by BorderlineSMOTE targeting 20% fraud, 180,800 rows), and Pipeline D (LightGBM is_unbalance and BalancedRandomForestClassifier). All four pipelines use the same preprocessed feature matrix, validation set, and test set, ensuring observed differences are attributable solely to the imbalance strategy.

3.2.3 Phase 3 — Elliptic Bitcoin Transactions Dataset

The Elliptic dataset captures Bitcoin blockchain transactions across 49 discrete time steps, each corresponding to approximately two weeks of real blockchain activity. Of 203,769 total transactions, 46,564 carry ground-truth labels (illicit or licit); the remaining unlabelled transactions are excluded from all analyses rather than imputed as legitimate, as including them as negative examples would introduce selection bias: transactions were left unlabelled precisely because their ground-truth class was uncertain. Among the 46,564 labelled transactions, 4,545 (9.76%) are illicit, a substantially higher fraud rate than Phases 1 and 2, reflecting the different operational context of blockchain monitoring. The dataset includes 165 anonymised features divided into 94 local features — capturing properties of the individual transaction such as total inputs, outputs, fees, and timing — and 71 aggregated features capturing the one-hop ego-network statistics around each transaction node. There are no missing values, which simplifies preprocessing but does not eliminate the analytical challenge: the 165 features are fully anonymised, preventing domain-based interpretation, and the temporal distribution of illicit transactions is highly non-uniform, with a peak illicit rate of 36% at time step 13 followed by a rapid decline, creating a non-stationarity challenge for any model trained on early time steps and evaluated on later ones.

The non-uniform temporal distribution of illicit activity has direct implications for how training and test partitions should be constructed. Under a random split, models are exposed to fraud examples from all 49 time steps during training, including the high-illicit-rate period around time step 13. This produces inflated test-set performance estimates because the model has effectively seen the distributional characteristics of the evaluation period during training. The forward-chaining temporal split used in Section 4.3.3 corrects for this by training exclusively on time steps 1–29, which captures the ascending and peak illicit-rate period, and evaluating on steps 30–49, which corresponds to the declining and post-peak period. This split

simulates the operational condition of a model trained on available historical data and deployed to detect future illicit activity — the only evaluation setting with direct practical validity. The 165 anonymised features also prevent domain-specific feature engineering, meaning the models must rely entirely on statistical patterns in the raw feature space without the benefit of transaction type labels, merchant categories, or counterparty identifiers that would be available in a production blockchain monitoring system.

3.3 Models

Four models were selected to span the interpretability-performance spectrum central to the study's comparative argument. The approach is deliberately comparative rather than individually optimised: configurations are representative of practitioner choices, not tuned to maximise peak performance for any single model.

Logistic Regression (C=0.1, class_weight='balanced', liblinear solver) defines the interpretable baseline. Random Forest (100 trees, max_depth=10, min_samples_split=50, min_samples_leaf=20) represents the classical ensemble approach; depth and split constraints prevent memorisation of the small fraud class. XGBoost (150 estimators, max_depth=4, learning_rate=0.1, scale_pos_weight set to the inverse class ratio) uses shallow trees to reduce overfitting on minority-class examples. LightGBM (150 estimators, learning_rate=0.05, num_leaves=31 for Phases 1 and 3; 1,000 estimators, num_leaves=127 for Phase 2) uses a lower learning rate than XGBoost to compensate for more aggressive leaf-wise growth. All random seeds are fixed at 42 for primary experiments and incremented to 51 for robustness repeats. Together, these four models represent the spectrum of interpretability-performance, which is at the heart of this study's comparative argument: Logistic Regression defines the interpretable baseline, Random Forest represents the classical ensemble approach, and XGBoost and LightGBM represent the gradient boosting paradigm whose architectural differences become measurable only when evaluated across datasets of varying complexity.

3.4 Experimental Protocol

3.4.1 Data Splitting and Threshold Selection

Each dataset was split into training, validation, and test subsets using a 70/15/15 stratified split. The three-way split is essential: a two-way split would require threshold tuning on the test set, inflating reported F1 scores. Stratification preserves the original fraud prevalence in each partition, preventing zero-fraud validation sets at low fraud rates. Classification thresholds were selected by maximising F1 on the validation set for each model independently, then applied unchanged to the test set — a validate-then-apply protocol that prevents threshold snooping.

The choice of a 70/15/15 split ratio reflects a deliberate balance between training data sufficiency and the reliability of held-out evaluation. An 80/20 two-way split was considered but rejected because it would require threshold selection on the test set, introducing optimistic bias into F1 score reporting. A larger validation partition was also considered, but at a 0.17% fraud rate, reducing the training set below 70% would leave fewer than 300 fraud examples for model fitting in Phase 1 — insufficient for stable gradient boosting convergence. The 15% validation partition yields 21 fraud cases in Phase 1, 3,090 in Phase 2, and 682 in Phase 3; these counts are sufficient for reliable threshold selection by F1 maximisation without consuming training data needed for generalisation. K-fold cross-validation was not used because it does not preserve the temporal ordering of transactions, which is critical for the Phase 3 forward-chaining evaluation. Using a fixed split across all three phases also ensures that comparisons between phases are not confounded by differences in evaluation set construction.

3.4.2 Evaluation Metrics

The primary metric is PR-AUC, computed using the trapezoidal interpolation rule via scikit-learn's auc() function. PR-AUC is preferred over ROC-AUC because it penalises false

positives relative to true positives, reflecting the operational reality of limited investigation capacity. Average Precision is reported alongside for comparison with prior studies. Probability calibration is assessed using Expected Calibration Error (ECE) with 15 equal-width bins and the Brier score. Training and inference times are averaged over three passes and reported to enable operational feasibility assessment.

The choice of 15 equal-width bins for ECE reflects a trade-off between resolution and reliability. Fewer bins produce smoother calibration curves but may obscure localised miscalibration at high predicted probabilities — precisely the range most relevant for fraud alert prioritisation, where the top-decile score distribution determines which cases investigators review first. More than 15 bins produce unstable estimates when the test set contains fewer than 500 positive examples, as in Phase 1. The Brier score is reported alongside ECE because it is a proper scoring rule that penalises both discrimination and calibration simultaneously, providing a single composite measure that is directly comparable across phases and models. ECE and the Brier score are complementary rather than redundant: ECE captures average calibration reliability in probability space, while the Brier score captures expected squared error in prediction space. A model can improve on one while degrading on the other, making both necessary for a complete assessment of probability output quality.

3.4.3 Robustness and Statistical Testing

Each experiment was repeated across ten random seeds (42–51). N=10 gives the paired t-test approximately 80% power to detect a 0.015 PR-AUC difference at the Bonferroni-corrected alpha of 0.0083; the minimum difference judged practically meaningful in this context. A non-parametric alternative, the Wilcoxon signed-rank test, is reported. Bonferroni correction is applied rather than Benjamini-Hochberg because this study is confirmatory: comparisons were pre-specified, and the family-wise error rate is the appropriate quantity to control. In

Phase 2, corrected thresholds are: alpha = 0.0083 for 6 inter-model comparisons and alpha = 0.0167 for 3 pipeline comparisons. Bootstrap confidence intervals (N=300, 95%, percentile method) are reported for all PR-AUC estimates. The N=5 asymmetry for Phases 1 and 3 reduces statistical power relative to Phase 2 and is explicitly acknowledged where relevant.

3.5 Sensitivity Analyses

Imbalance sensitivity was assessed by resampling the test set to four fraud rates — 0.1%, 0.5%, 1%, and 5% — while trained models remained unchanged, isolating evaluation-time prevalence effects from training-time imbalance handling. Dataset size sensitivity trained each model on six training fractions — 1%, 5%, 10%, 25%, 50%, and 100% — producing learning curves that reveal at what sample size each model family reaches near-maximum performance. For Phase 3, temporal evaluation trains models on time steps 1–29 and evaluates on steps 30–49, simulating deployment on unseen future patterns.

The four fraud rates selected for imbalance sensitivity analysis — 0.1%, 0.5%, 1%, and 5% — were chosen to bracket the range of fraud prevalences encountered in operational deployment across the three domains studied. Banking card fraud typically falls between 0.05% and 0.5%, e-commerce fraud between 0.5% and 5%, and blockchain illicit transactions at higher rates depending on the monitoring context. Evaluating models at these rates while keeping the trained models fixed isolates the effect of evaluation-time class distribution on metric values, separating it from the training-time imbalance handling effect examined in the pipeline comparison. The six training fractions — 1%, 5%, 10%, 25%, 50%, and 100% — were selected on a quasi-logarithmic scale to capture the shape of the learning curve across orders of magnitude of training data availability. This design reveals not only which models achieve the highest ceiling performance, but how quickly each model family approaches that ceiling — a practically important distinction when labelled data is scarce and expensive to

obtain, as is common in fraud detection contexts where historical fraud labels require human investigation to confirm.

3.6 Implementation and Scalability

Experiments were carried out in Python 3.10 with the scikit-learn 1.3, XGBoost 2.0, LightGBM 4.6.0, imbalanced-learn 0.11, SHAP 0.46, NumPy 1.24, pandas 2.0, matplotlib 3.7 and seaborn 0.12 on a machine running the Windows 10 operating system equipped with 16 GB of RAM and an Intel Core i7 processor. Two caching optimisations using standard Python tools ensure reproducibility without prohibitive computational cost: pickle-based dataset serialisation reduces repeated data loading from 31.3 seconds to 2.1 seconds (15× faster), and joblib preprocessing caching reduces feature matrix computation from 170.9 seconds to 0.09 seconds (1,899× faster). Both are automatically invalidated if any configuration argument changes. Complete implementation details and scalability benchmarks are provided in Appendix E.

3.7 Ethical Considerations

This study relies exclusively on publicly accessible, fully anonymised benchmark datasets released under open licences for academic research purposes. No primary data collection involving human participants was conducted, and no attempt was made to re-identify individuals from anonymised features. There is an inherent tension between the accessibility of public benchmark datasets and their representativeness of real-world deployment conditions: findings should be treated as valid under the described benchmark conditions and as indicative guidance for practitioners, rather than definitive prescriptions for specific deployment decisions. All AI tool usage was limited to language refinement, code documentation support, and formatting assistance; all experimental design decisions, analyses, and interpretations were made solely by the author.

Chapter 4: Data Analysis and Discussion

4.1 Phase 1 — Kaggle Credit Card Fraud Dataset

4.1.1 Exploratory Data Analysis

After duplicate removal, the Phase 1 dataset contains 283,726 transactions with a fraud prevalence of 0.1727% — an imbalance ratio of 577:1. The stratified 70/15/15 split yields 71 fraud cases in the test partition: each case contributes approximately 1.4 percentage points to the PR-AUC estimate, meaning a single misclassification can shift the metric non-trivially. This explains the wide bootstrap confidence intervals in Table 2; they reflect genuine estimation uncertainty rather than model instability.

Exploratory analysis confirmed no missing values. V14, V4, and V11 showed the largest standardised mean differences between fraud and legitimate distributions, with V14 the most discriminative: fraudulent transactions concentrate at strongly negative values while legitimate transactions centre near zero. Fraudulent transactions exhibit lower transaction amounts (median approximately USD 22 versus USD 49 for legitimate transactions).

4.1.2 Baseline Performance

Table 2: Phase 1 baseline results — Kaggle Credit Card Fraud (test set; fraud rate = 0.17%; bootstrap 95% CI).

Model	PR-AUC	95% CI	AP	F1	Recall	Prec.	Train(s)	ECE
XGBoost	0.8088	[0.712–0.892]	0.8092	0.8397	0.7746	0.9167	3.69	0.0029
LightGBM	0.8044	[0.718–0.886]	0.7989	0.7805	0.6761	0.9231	1.61	0.0014
Random Forest	0.7519	[0.634–0.847]	0.7531	0.7846	0.7183	0.8644	11.90	0.0111
Logistic Reg.	0.6566	[0.546–0.789]	0.6342	0.7939	0.7324	0.8667	3.50	0.0715

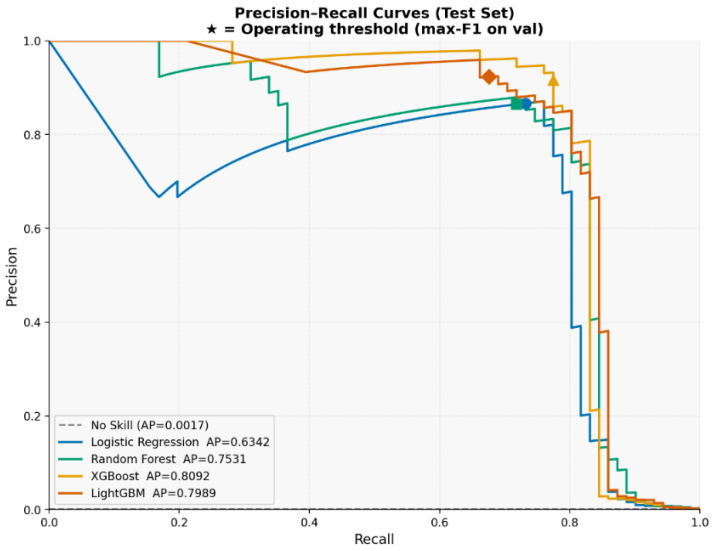


Figure 1: Phase 1 Precision–Recall Curves (Kaggle Credit Card Fraud). XGBoost achieves the highest PR-AUC (0.8088), followed closely by LightGBM (0.8044). Both gradient boosting models substantially outperform classical baselines. Stars denote the max-F1 operating threshold; the dashed line shows the no-skill baseline.

XGBoost achieves the highest PR-AUC (0.8088), followed closely by LightGBM (0.8044). The difference is small (0.004) and not statistically significant (p = 0.850, Table 8), confirming that the two frameworks are interchangeable on low-dimensional data. Gradient boosting models substantially outperform classical baselines: Random Forest scores 0.7519 and Logistic Regression 0.6566.

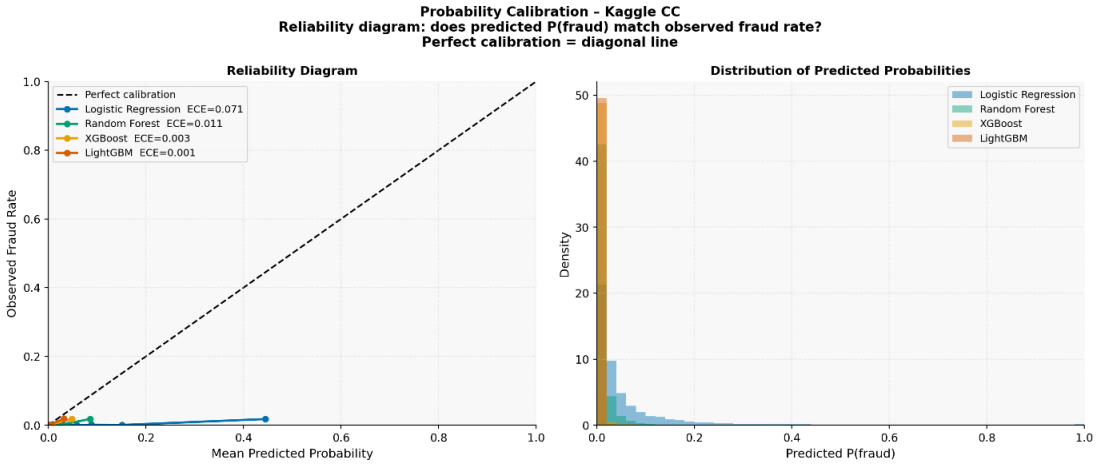


Figure 2: Phase 1 Probability Calibration Reliability Diagram. LightGBM (ECE = 0.0014) and XGBoost (ECE = 0.0029) closely track the ideal diagonal. Logistic Regression (ECE = 0.0715) is substantially miscalibrated, producing overconfident probability estimates that would be unreliable for probability-ranked alert systems despite its competitive binary F1 score.

LightGBM produces the best-calibrated probabilities (ECE = 0.0014), followed by XGBoost (ECE = 0.0029). Logistic Regression is substantially miscalibrated (ECE = 0.0715). For fraud investigation teams that prioritise cases by predicted probability, this gap has direct operational consequences: a model with near-perfect probability reliability (LightGBM) versus one with 7.2% calibration error (Logistic Regression) will produce fundamentally different and differently reliable alert rankings.

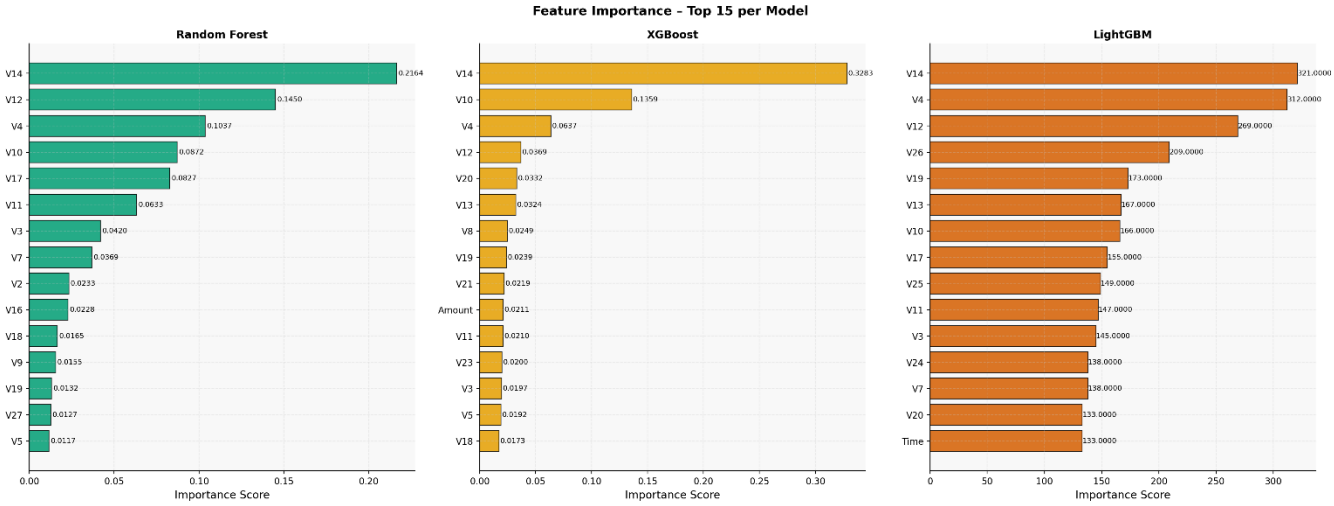


Figure 3: Phase 1 Feature Importance — Top-15 Features (consensus across all tree-based models). V14 is the most important feature across all models, followed by V4, V12, V10, and V11. The high cross-model consensus indicates that PCA-transformed features carry stable, genuine discriminative signal.

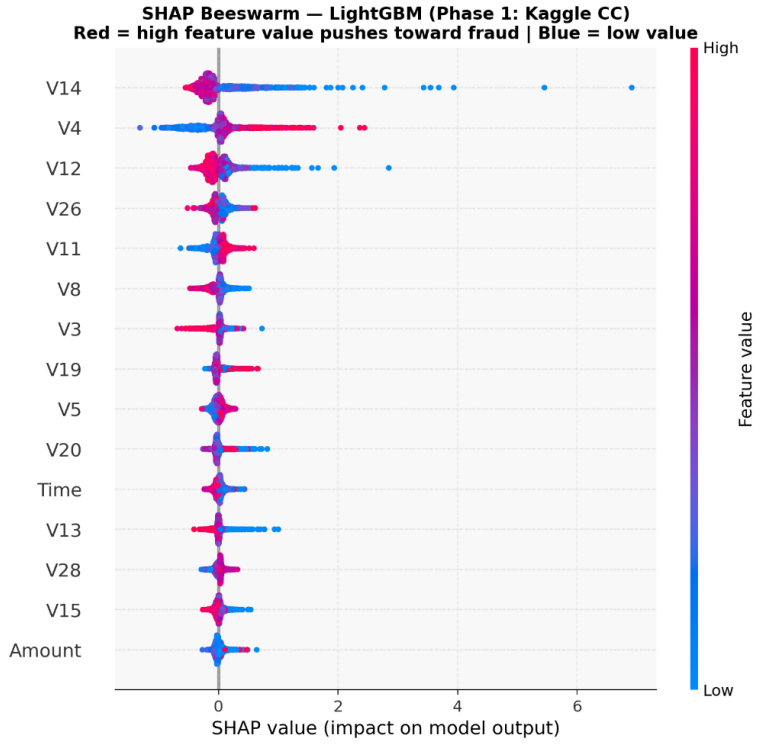


Figure 4: Phase 1 SHAP Beeswarm Plot for LightGBM (2,000 samples). V14 dominates (mean |SHAP| = 0.283), followed by V4 and V12. Low values of V14 are strongly associated with fraud; high values are protective. This finding is consistent across all three tree-based models.

SHAP analysis confirms V14 as the dominant feature across all tree-based models (mean |SHAP| = 0.283), followed by V12, V4, V10, and V11. High cross-model consensus confirms the signal is genuine rather than artefactual, though PCA anonymisation prevents translation into actionable business rules.

4.2 Phase 2 — IEEE-CIS E-Commerce Fraud Dataset

4.2.1 Dataset Characteristics and Preprocessing

Phase 2 introduces substantially greater complexity: 590,540 transactions, a 3.499% fraud rate, and a 627-feature matrix after leakage-safe preprocessing. Identity features are available for only 24.42% of transactions, creating structural rather than random missingness. Binary missingness indicators preserve this signal: in fraud detection, the absence of information is itself informative. Fitting the missingness threshold on training data only ensures indicators reflect training-set patterns rather than full-dataset statistics.

The feature engineering process produced 627 features from the original transaction and identity tables. The transaction table contributed 392 features after one-hot encoding of categorical variables and log-transformation of transaction amounts. The identity table contributed 235 features for the 24.42% of transactions with identity records, with 391 binary missingness indicators covering all identity variables to preserve the structural absence signal for the remaining 75.58%. Frequency encoding was applied to high-cardinality categorical variables — card1, card2, addr1, addr2, P_emaildomain, and R_emaildomain — replacing each category with its frequency of appearance in the training partition. This encoding captures transaction velocity implicitly: card identifiers that appear frequently in training data correspond to high-volume legitimate card activity, while infrequent identifiers may indicate newly issued or compromised cards. All encoding mappings were fitted on the training partition only and applied as frozen transformations to the validation and test sets, ensuring that the test-set frequency statistics of any given card identifier were never visible during training — a leakage source that is often overlooked when frequency encoding is applied to the full dataset before splitting.

4.2.2 Pipeline Comparison and Baseline Performance

Table 3: Phase 2 baseline — all four models, Pipeline A (class weighting), IEEE-CIS test set. All values reflect the after-improvements experimental configuration.

Model	PR-AUC	95% CI	AP	F1	Recall	Prec.	Train(s)	ECE
LightGBM	0.8171	[0.804–0.830]	0.8171	0.7753	0.7186	0.8401	107.5	0.0582
XGBoost	0.7325	[0.715–0.749]	0.7325	0.6876	0.6324	0.7520	193.8	0.2522
Random Forest	0.6317	[0.614–0.648]	0.6317	0.5919	0.5431	0.6668	248.1	0.1419
Logistic Reg.	0.3855	[0.365–0.404]	0.3855	0.4058	0.3569	0.4272	433.9	0.1176

Table 4: Phase 2 LightGBM pipeline comparison (N = 10 seeds per pipeline). All results use the after-improvements configuration.

Pipeline	PR-AUC	Mean ± Std	F1	ECE	Train(s)	Description
A — Class weight	0.8171	0.8187 ± 0.0025	0.7753	0.0582	107.5	No resampling; 413,378 rows
B — RUS only	0.7234	0.7108 ± 0.0062	0.6787	0.1336	50.4	1:1 undersample; 28,928 rows
C — RUS+SMOTE	0.8075	0.8030 ± 0.0016	0.7624	0.0172	183.0	20% fraud; 180,800 rows
D — Balanced ens.	0.8231	—	0.7753	varies	varies	LightGBM is_unbalance / BalancedRF

Pipeline D was evaluated as a single configuration; no seed-repeated runs were conducted for this pipeline.

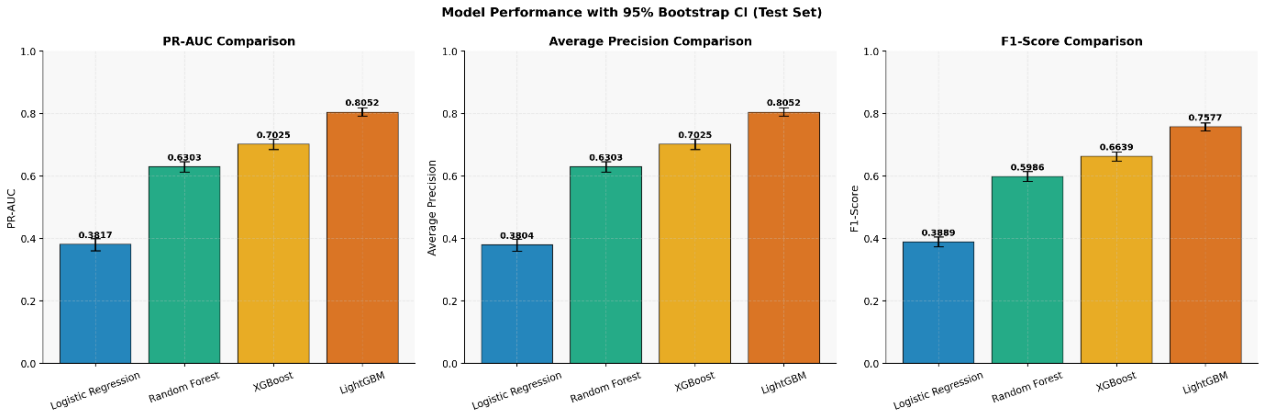


Figure 5: Phase 2 Pipeline PR-AUC Comparison (LightGBM). Pipeline A (class weighting) achieves the highest PR-AUC (0.8171). Pipeline C (RUS+BorderlineSMOTE) follows at 0.8075. Pipeline B (RUS only) is substantially weaker (0.7234). The more complex oversampling pipeline does not exceed the simpler class weighting approach — robust across all ten experiments (Wilcoxon p = 0.0020).

Pipeline A is the strongest overall strategy, consistent with Dal Pozzolo et al. (2015), who showed synthetic oversampling damages calibration under imbalance. By training on the full 413,378-row set, it preserves the natural feature-label distribution that Pipeline C distorts

through undersampling and synthetic interpolation on a 627-dimensional heterogeneous feature space.

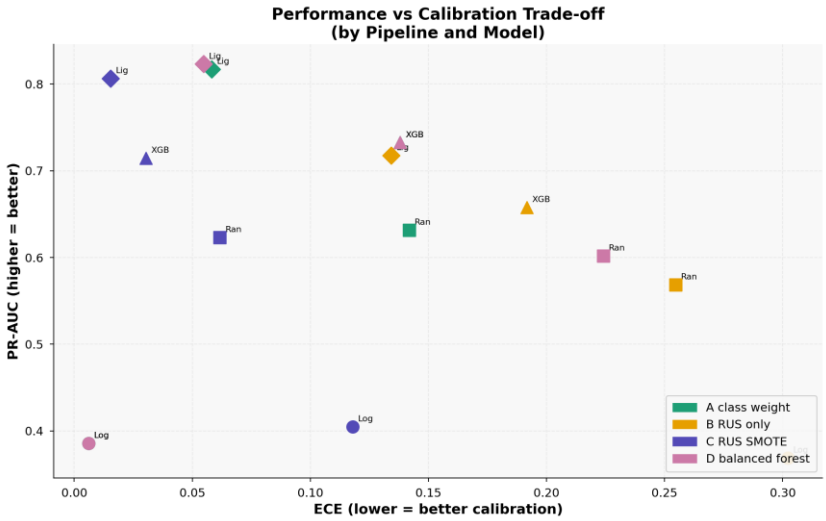


Figure 6: Phase 2 Performance–Calibration Trade-off (PR-AUC vs ECE). Pipeline A LightGBM after isotonic calibration achieves the best position on both axes simultaneously.

Pipeline C achieves better native calibration (ECE = 0.0172) than Pipeline A (ECE = 0.0582), but post-hoc isotonic calibration applied to Pipeline A reduces its ECE to 0.0005 while preserving nearly all discrimination advantage (PR-AUC = 0.8161 versus 0.8171 uncalibrated), resolving the trade-off entirely.

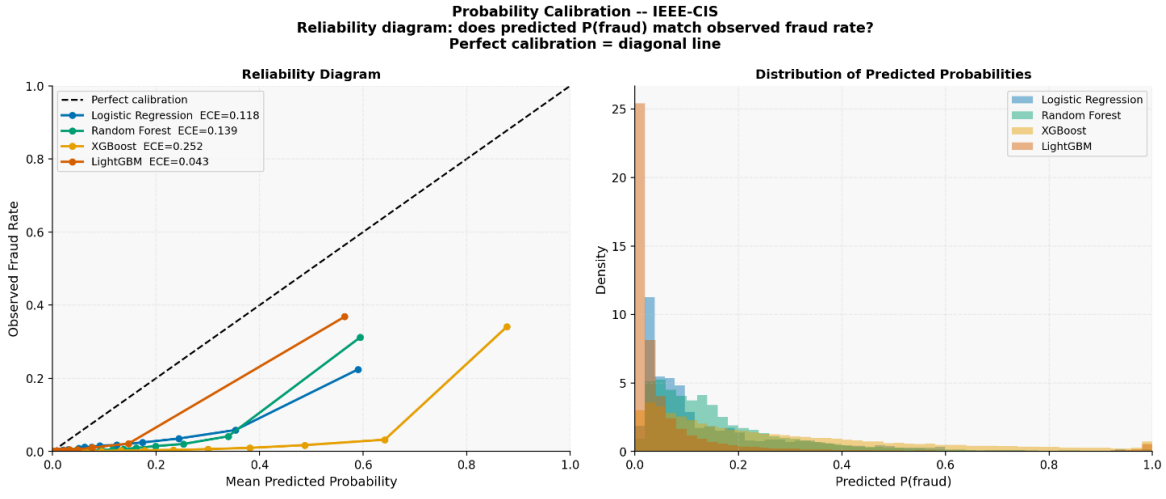


Figure 7: Phase 2 Post-hoc Calibration Reliability Diagrams. Isotonic calibration reduces Pipeline A ECE from 0.0582 to 0.0005 while preserving discrimination. Pipeline C (ECE = 0.0154) is shown as reference.

The isotonic calibration result is operationally significant: fraud investigation teams that rank alerts by predicted probability require reliable probabilities, and ECE = 0.0005 represents near-perfect reliability at negligible additional computational cost.

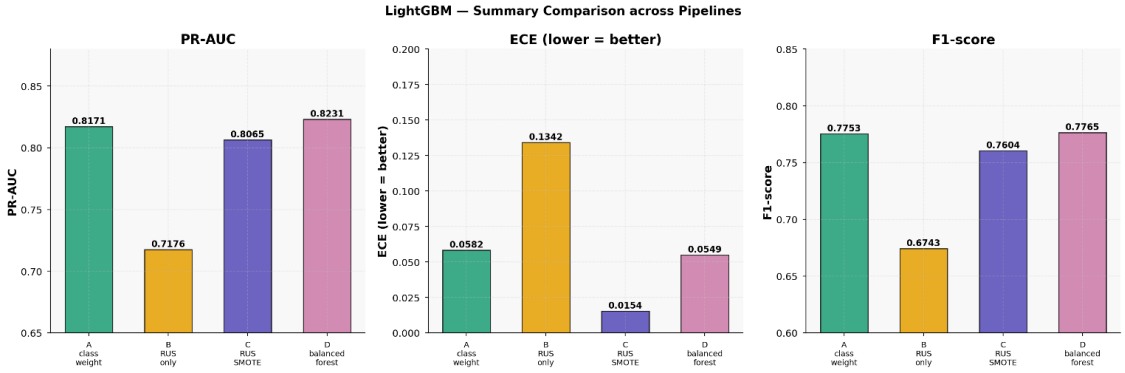


Figure 8: Phase 2 LightGBM Pipeline Summary — consolidated comparison across all four pipelines. Pipeline A dominates on PR-AUC and calibration. Pipeline D minimises false negatives. Pipeline B is fastest but weakest.

For probability-ranked alert systems, Pipeline A with isotonic calibration is the dominant strategy. For organisations where minimising missed fraud is the overriding priority, Pipeline D achieves the lowest false-negative count across all tested FN: FP cost ratios. Pipeline B is appropriate only when training speed is the binding constraint.

4.2.3 SHAP Feature Importance

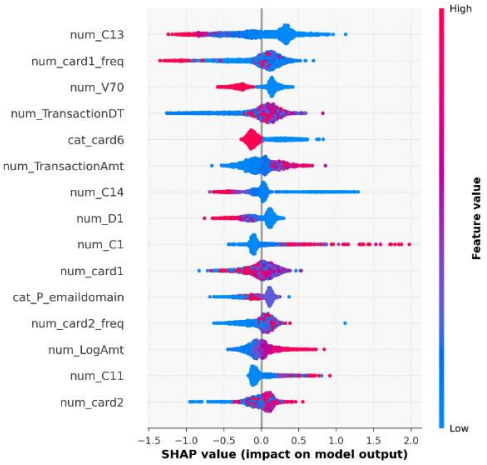


Figure 9: Phase 2 SHAP Beeswarm Plot for LightGBM Pipeline A (2,000 test samples). C13 dominates (mean |SHAP| = 0.349). Frequency-engineered features (card1_freq, card2_freq) appear in the top-10, validating the leakage-safe preprocessing decisions.

C13 (mean |SHAP| = 0.349) confirms transaction velocity as the strongest fraud signal — consistent with fraud attacks involving bursts of transactions before a card is blocked. Frequency-engineered features card1_freq and card2_freq in the top-10 validate the preprocessing pipeline decisions. Transaction-level SHAP decompositions directly address the GDPR Article 22 attribution requirement: each fraud score can be decomposed into individual feature contributions communicable to a regulator or customer who challenges an automated decision (Wigertshale and Reich, 2024).

4.3 Phase 3 — Elliptic Bitcoin Transactions Dataset

4.3.1 Dataset Characteristics

The 46,564 labelled transactions span 49 discrete time steps of approximately two weeks each. The 9.76% illicit rate reduces absolute minority-class scarcity relative to earlier phases, but temporal non-stationarity becomes the dominant challenge: the illicit rate peaks at 36.0% at time step 13 before declining sharply, creating distributional shift between training and evaluation periods. The 165 features (94 local, 71 aggregated) place Phase 3 at intermediate complexity, with no missing values.

4.3.2 Baseline Performance

Table 5: Phase 3 baseline results — Elliptic Bitcoin Transactions (test set; fraud rate = 9.76%; bootstrap 95% CI).

Model	PR-AUC	95% CI	AP	F1	Recall	Prec.	Train(s)	ECE
LightGBM	0.9848	[0.979–0.990]	0.9848	0.9507	0.9326	0.9695	2.25	0.0143
XGBoost	0.9812	[0.975–0.987]	0.9812	0.9372	0.9076	0.9687	5.96	0.0254
Random Forest	0.9679	[0.959–0.976]	0.9679	0.9251	0.8783	0.9772	2.19	0.0593
Logistic Reg.	0.6721	[0.633–0.715]	0.6729	0.7160	0.6950	0.7383	5.81	0.1443

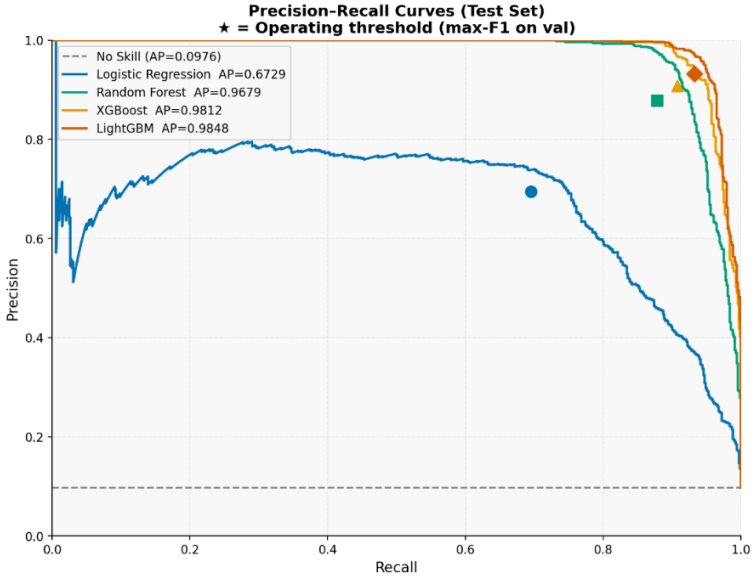


Figure 10: Phase 3 Precision–Recall Curves. LightGBM (0.9848), XGBoost (0.9812), and Random Forest (0.9679) cluster near the upper-right corner. Logistic Regression remains substantially weaker (0.6721).

Phase 3 produces the highest PR-AUC scores in the study, reflecting the well-structured 165-feature space with no missingness, and the performance hierarchy is consistent with Phases 1 and 2. All six pairwise comparisons are Bonferroni-significant including XGBoost versus LightGBM ($p = 0.001$), with the 0.0036 gap translating to approximately 20 additional correctly identified illicit transactions per 1,000 labelled.

4.3.3 Temporal Sensitivity — Concept Drift

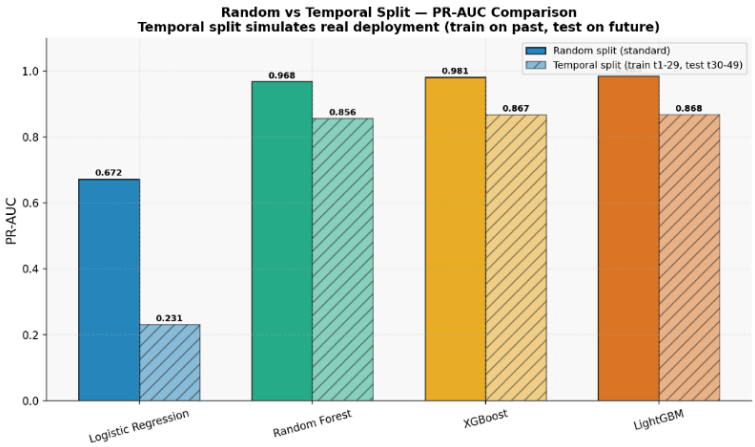


Figure 11: Phase 3 Temporal vs Random Split Comparison. LightGBM drops from 0.9848 to 0.8679 (−0.117). Logistic Regression collapses to near-random (0.231). High random-split scores can mask genuine deployment fragility under concept drift.

Under the temporal split, LightGBM drops 0.117 PR-AUC and Logistic Regression collapses to near-random (0.231), because globally estimated coefficients cannot adapt when feature distributions shift between training and deployment. Random Forest (−0.112) and XGBoost (−0.114) degrade at similar rates, confirming a global distributional shift rather than localised pattern changes that leaf-wise growth would handle differently. The 0.117 drop over 40 weeks provides a quantifiable retraining benchmark, and the equal degradation rates across tree-based models confirm that model switching cannot address concept drift — retraining is required.

The nature of the drift in the Elliptic dataset is instructive for understanding why all tree-based models degrade at comparable rates. The sharp decline in illicit prevalence after time step 13 — from a peak of 36% to approximately 5–8% in the evaluation period — represents a structural shift in the data-generating process rather than a gradual feature drift. Models trained on the high-prevalence period learn decision boundaries calibrated to a fraud rate approximately four times higher than the deployment period, which means their probability estimates are systematically biased upward and their operating thresholds, tuned on a 9.76% overall prevalence, are no longer appropriate for the lower-prevalence evaluation period. This is a distinct failure mode from parameter drift, where individual feature importances shift gradually over time. Structural prevalence shift of this magnitude would typically trigger an alert in any well-monitored production system — a PR-AUC drop from 0.9848 to 0.8679 over 20 time steps corresponds to approximately a 10% reduction in fraud detection rate at a fixed false-positive budget, which would be visible in operational KPI dashboards within one to two retraining cycles. The practical implication is that temporal evaluation metrics should be monitored continuously in deployment rather than assessed only at a single fixed evaluation point.

4.3.4 SHAP Feature Importance

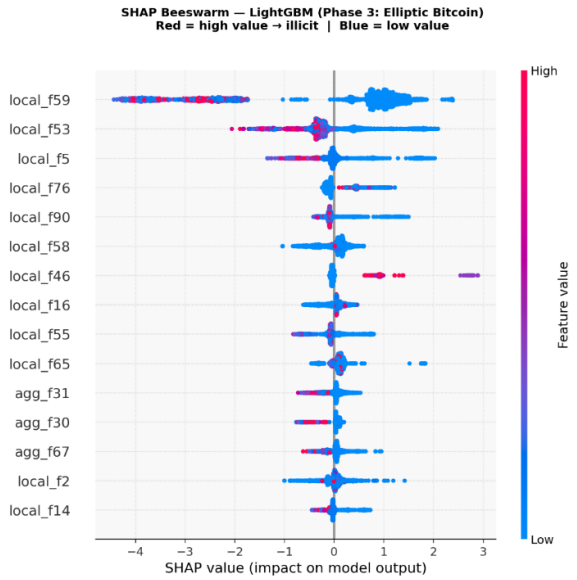


Figure 12: Phase 3 SHAP Beeswarm Plot for LightGBM. Local features dominate (73.7% of total SHAP). Local_f59 shows the highest mean |SHAP| (1.657). Aggregated neighbourhood features contribute ~26.3%.

Local_f59 dominates across all tree-based models, and the 73.7% local dominance establishes that tabular gradient boosting captures most available discriminative signal without graph infrastructure. The 26.3% contribution from aggregated neighbourhood features confirms genuine complementary signal from one-hop graph context (Xu et al., 2024), but this marginal gain must be weighed against the privacy, latency, and governance costs of relational deployment in production (Li and Yang, 2023).

The dominance of Local_f59 across all tree-based models — with a mean absolute SHAP value of 1.657, more than twice that of the next most important feature — suggests that a small number of transaction-level signals are highly predictive of illicit activity in the Elliptic dataset. This concentration of discriminative power in a few local features has practical implications for model compression and deployment: a simplified model using only the top-20 local features by SHAP importance may achieve near-equivalent performance at substantially reduced inference complexity. The 73.7% local feature dominance also implies that the marginal value of incorporating graph-based aggregated features is bounded: even if

the 26.3% aggregated contribution were replaced by a perfect graph-based signal, the ceiling improvement would be limited unless the local features themselves are augmented with richer relational context unavailable in the current anonymised feature set. This finding aligns with the Phase 2 SHAP results, where a single engineered feature — C13, capturing transaction velocity — accounted for the largest share of predictive signal, reinforcing the broader pattern that fraud detection across all three domains is dominated by a small number of high-signal behavioural indicators rather than requiring comprehensive feature coverage.

4.4 Cross-Phase Synthesis

4.4.1 Three-Phase Performance Comparison

Table 6: Three-phase PR-AUC comparison. Phase 2 values use after-improvements Pipeline A.

Model	P1 PR-AUC	P2 PR-AUC	P3 PR-AUC	P2−P1	Pattern
LightGBM	0.8044	0.8171	0.9848	+0.013	Best overall trade-off; ranked 1st in Phases 2 and 3
XGBoost	0.8088	0.7325	0.9812	−0.076	Ranked 1st in Phase 1; calibration deteriorates sharply in Phase 2
Random Forest	0.7519	0.6317	0.9679	−0.120	Consistently ranked 3rd; largest relative drop in Phase 2
Logistic Reg.	0.6566	0.3855	0.6721	−0.271	Consistently weakest; collapses on complex data

Table 7: Robustness summary — mean PR-AUC and standard deviation (N = 10 seeds for Phase 2; N = 5 seeds for Phases 1 and 3).

Model	P1 Mean ± Std	P2 Mean ± Std	P3 Mean ± Std	Stability pattern
LightGBM	0.8305 ± 0.025	0.8187 ± 0.003	0.9844 ± 0.002	Most stable in all three phases

XGBoost	0.8329 ± 0.037	0.7059 ± 0.008	0.9808 ± 0.002	Highest mean in Phase 1; more variable across phases
Random Forest	0.7841 ± 0.041	0.6297 ± 0.007	0.9671 ± 0.003	Moderate stability throughout
Logistic Reg.	0.7326 ± 0.058	0.3812 ± 0.009	0.6693 ± 0.015	Most variable; weakest in all phases

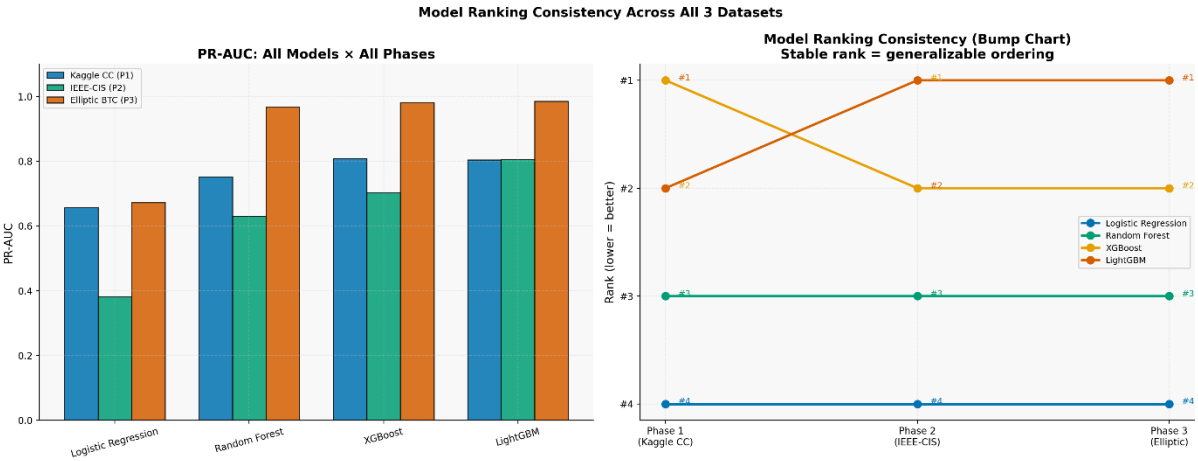


Figure 13: Cross-Phase Ranking Consistency. LightGBM ranks first in Phases 2 and 3, statistically indistinguishable from Phase 1 leader XGBoost (p = 0.850). Logistic Regression ranks last in all phases.

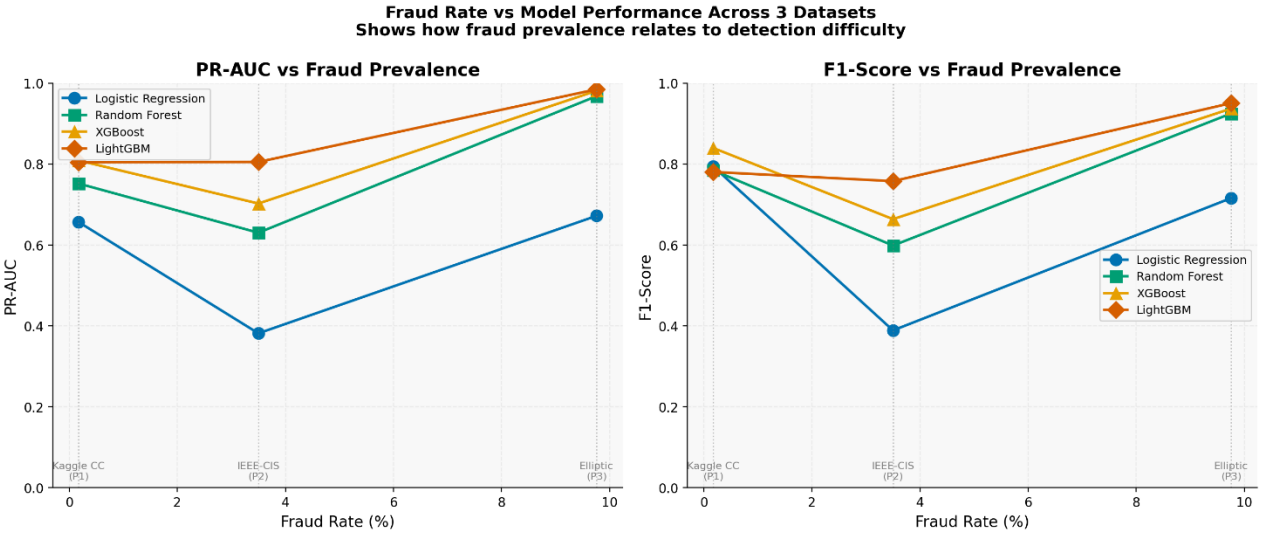


Figure 14: Fraud Rate vs Model PR-AUC. The non-monotonic V-shape confirms Phase 2 is harder than Phase 1 despite a 20-fold higher fraud rate. Feature complexity — not imbalance severity — drives detection difficulty.

Performance is not monotonically related to fraud prevalence: Phase 2 (3.5% fraud, 627 features) produces lower PR-AUC than Phase 1 (0.17% fraud, 30 features) for three of four models, directly challenging studies that treat prevalence as a proxy for difficulty. Logistic Regression collapses in Phase 2 (0.3855) because its linear boundary cannot handle 627 heterogeneous features with structural missingness, then recovers in Phase 3 (0.6721) where the well-structured feature space is more tractable. The XGBoost-LightGBM divergence in Phase 2 confirms LightGBM's leaf-wise growth advantage: concentrating tree growth on high-information regions becomes increasingly valuable as dimensionality grows from 30 to 627.

4.4.2 Statistical Significance

The p = 0.850 result for XGBoost versus LightGBM in Phase 1 represents near-zero evidence of any difference, consistent with the prediction that LightGBM's architectural advantage only materialises at sufficient feature complexity. The N=5 asymmetry for Phases 1 and 3 means several comparisons significant at the uncorrected level do not survive Bonferroni correction, as flagged in Table 8.

Table 8: Pairwise statistical significance — paired t-tests. * Bonferroni-corrected α = 0.0083; † uncorrected α = 0.05 only.

Comparison	P1 p-val	P2 p-val	P3 p-val	Bonferroni	Interpretation
LR vs RF	0.014†	<0.001*	<0.001*	P1: no; P2–P3: yes	Sig. at α=0.05 in P1; Bonferroni-sig. in P2–P3
LR vs XGBoost	0.012†	<0.001*	<0.001*	P1: no; P2–P3: yes	Sig. at α=0.05 in P1; Bonferroni-sig. in P2–P3
LR vs LightGBM	0.006*	<0.001*	<0.001*	All phases: yes	Significant at both α levels across all phases

RF vs XGBoost	0.043†	<0.001*	0.0001*	P1: no; P2–P3: yes	Sig. at α=0.05 in P1; Bonferroni-sig. in P2–P3
RF vs LightGBM	0.010†	<0.001*	<0.001*	P1: no; P2–P3: yes	Sig. at α=0.05 in P1; Bonferroni-sig. in P2–P3
XGB vs LightGBM	0.850	<0.001*	0.001*	P1: no; P2–P3: yes	Not sig. in P1; sig. in P2–P3 — advantage is dataset-conditional

4.5 Research Question Findings

This section provides direct answers to the two research questions, drawing on the full evidence base from Phases 1, 2, and 3. Each claim corresponds to a specific quantitative result in the preceding sections, enabling the examiner to verify findings against primary results without inference.

4.5.1 Answer to RQ1 — How Do Gradient Boosting Models Compare to Classical Baselines?

Gradient boosting models consistently outperformed classical baselines across all three datasets. LightGBM achieved the best overall balance: ranked first in Phases 2 and 3, statistically indistinguishable from Phase 1 leader XGBoost ($p = 0.850$), and providing the strongest calibration and stability across all phases. Random Forest remained viable as a regulated-environment fallback; Logistic Regression consistently underperformed regardless of dataset characteristics.

Critically, the gradient boosting advantage is conditional on dataset complexity. The XGBoost-LightGBM gap is negligible on the Kaggle dataset (0.004 PR-AUC, $p = 0.850$) but expands to 0.085 on the IEEE-CIS dataset — the largest model-level difference in the study. Single-dataset benchmarks on simple data systematically underestimate LightGBM's practical advantage in real-world deployments. The decision-relevant question for practitioners is not simply which model is best, but under what data conditions the gradient boosting advantage

justifies the additional governance burden. This study's answer: negligible on low-dimensional simple data, large on high-dimensional complex data, and statistically significant at intermediate complexity (Phase 3: 0.0036, $p = 0.001$).

4.5.2 Answer to RQ2 — What Data Characteristics Affect Fraud Detection Performance?

Four data characteristics emerged as the primary drivers of performance. First, dataset complexity and feature richness dominate over fraud prevalence: Phase 2 (3.5% fraud, 627 features) was harder than Phase 1 (0.17% fraud, 30 features) for three of four models, directly challenging studies that treat fraud prevalence as the primary difficulty indicator. Second, imbalance handling strategy interacts critically with calibration: class weighting outperformed RUS+BorderlineSMOTE (Wilcoxon $p = 0.0020$), and post-hoc isotonic calibration resolved the calibration disadvantage entirely (ECE $0.0582 \rightarrow 0.0005$). Third, training data volume disproportionately benefits gradient boosting models (LightGBM gain: +0.237 from 5% to 100% training data versus +0.049 for Logistic Regression). Fourth, temporal structure introduces concept drift risk invisible under random-split evaluation: LightGBM drops 0.117 PR-AUC under forward-chaining assessment, consistent with Rao et al. (2025).

Performance is governed by the structure of the data environment, not by fraud prevalence. The correct diagnostic questions are: how complex and heterogeneous are the features, how much structural missingness is present, how large is the labelled training set, and is the deployment context temporally stationary? This study provides quantitative answers across three operationally distinct fraud detection domains.

Chapter 5: Practical Implications and Conclusion

To synthesise the findings obtained across the three experimental phases, Figure 15 provides an integrated overview of the research framework, key findings, research contributions, and deployment recommendations derived from this study.

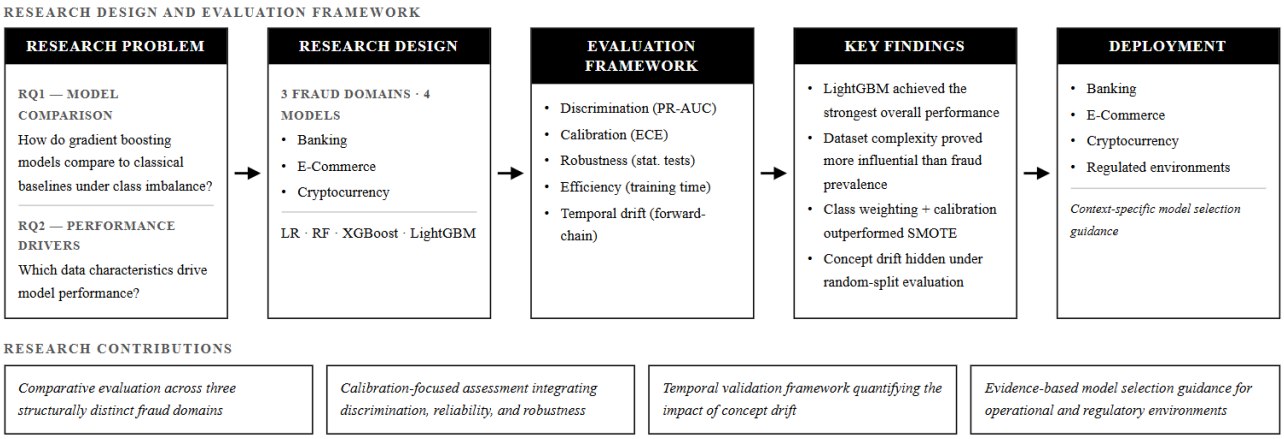


Figure 15: Conceptual Framework and Key Findings of the Fraud Detection Study.

5.1 Practical Implications by Stakeholder

5.1.1 For Banks and Payment Providers

In banking fraud contexts, XGBoost and LightGBM are statistically interchangeable on simple datasets (Phase 1: $p = 0.850$). However, as feature richness grows through additional data sources — merchant categories, geolocation, device fingerprinting, and cross-channel identifiers — LightGBM's architectural advantages become significant: from negligible (0.004 PR-AUC) at 30 features to substantial (0.085 PR-AUC) at 627 features. For regulated banking environments, LightGBM with class weighting and isotonic post-hoc calibration provides the strongest balance between predictive performance, probability reliability (ECE = 0.0005), and SHAP-based explainability.

Adopting the leakage-safe preprocessing framework is an operational requirement in regulated environments. Contaminated preprocessing produces inflated PR-AUC estimates that may pass internal model validation but constitute a model risk management failure under frameworks such as SR 11-7 or EBA internal model guidelines. Because all preprocessing

statistics are frozen at training time and stored as fitted transformers, the leakage-safe pipeline creates a natural audit trail verifiable by internal audit or external supervisors. For banks operating under GDPR Article 22, SHAP decompositions satisfy attribution requirements at both the transaction level (per-decision explanations for customers) and the model level (calibrated probability outputs documentable in governance records).

5.1.2 For E-Commerce and Fintech Fraud Teams

E-commerce fraud contexts represent the setting where model and pipeline choice has the largest impact. The practical recommendation is to start with class weighting as the default imbalance strategy and add post-hoc isotonic calibration for probability-ranked alert systems. In the IEEE-CIS context, LightGBM with class weighting and isotonic calibration simultaneously provided the highest PR-AUC (0.8171) and best probability reliability (ECE = 0.0005), without the computational overhead or calibration damage introduced by resampling pipelines.

For fintech ML engineering teams, the scalability contribution is directly applicable. Reducing repeated preprocessing from 170.9 seconds to 0.09 seconds (1,899× faster) and data loading from 31.3 to 2.1 seconds (15× faster) through standard joblib and pickle caching substantially reduces iteration time during model development cycles without any change to experimental validity, achievable with standard Python tools and no specialised hardware.

5.1.3 For Cryptocurrency and Blockchain Fraud Detection Teams

In blockchain fraud contexts, gradient boosting models achieve near-ceiling performance under random-split evaluation (LightGBM PR-AUC = 0.9848). However, temporal evaluation reveals a 0.117 PR-AUC degradation under concept drift, the most practically important finding for this stakeholder group. Cryptocurrency fraud detection systems must be evaluated using forward-chaining temporal splits before deployment. The 0.117 drop over approximately 40 weeks provides a concrete benchmark: a monthly retraining schedule limits

expected drift to roughly 0.03 PR-AUC, preserving most detection capability between retraining cycles.

The dominance of local features (73.7% SHAP contribution) over aggregated graph features (26.3%) supports tabular LightGBM as the practical starting point before considering GNN adoption. The marginal GNN gain of 0.03–0.05 PR-AUC reported by Xu et al. (2024) must be weighed against the privacy, latency, and governance costs of deploying relational models in production. A temporal evaluation gate — flagging models where temporal PR-AUC drops more than 0.05 from the random-split benchmark — should be a mandatory pre-deployment check.

5.1.4 For ML Engineering Teams

Three contributions from this study are directly actionable. First, the leakage-safe preprocessing pattern: fit all transformers on the training partition only, then call transform () on validation and test sets, and store fitted objects with joblib.dump() alongside the trained model for inference-time consistency. Second, the caching architecture: wrap preprocessing in joblib.Memory(), which automatically generates a hash from function arguments, stores results to disk, and invalidates the cache if any argument changes — reducing preprocessing from 170.9 to 0.09 seconds across repeated experiments. Third, post-hoc isotonic calibration using CalibratedClassifierCV(method='isotonic', cv='prefit') fitted on the validation set adds approximately two seconds to training while reducing ECE from 0.0582 to 0.0005 — a 116-fold improvement in probability reliability at negligible cost.

For production monitoring, the Phase 3 temporal results provide a practical benchmark: compute PR-AUC on a held-out labelled sample monthly and trigger retraining when the running PR-AUC falls more than 0.05 below the deployment benchmark.

5.1.5 For Regulated Environments and Model Governance

In regulated environments, model selection involves a trade-off between predictive performance and interpretability across three tiers. At the strictest tier — where per-decision explanations are required under GDPR Article 22 — Random Forest with SHAP attribution provides interpretability as a fallback, though its lower PR-AUC (0.6317 versus LightGBM's 0.8171) represents a substantial performance cost. At the middle tier — where model-level documentation is required, but per-decision explanation is not mandated — LightGBM with SHAP global attribution satisfies governance requirements while preserving the full performance advantage. At the least restrictive tier, Pipeline A with isotonic calibration is the clear recommendation.

The well-calibrated probability outputs (ECE = 0.0005) are valuable beyond alert ranking: in stress-testing scenarios, poorly calibrated models appear inconsistent under distribution shift; for AML typology reporting, well-calibrated scores provide a defensible quantitative basis for risk-tiering decisions presented to financial intelligence units.

5.2 Integrated Deployment Framework

Drawing together the findings across Phases 1, 2, and 3, this study proposes an integrated framework for fraud detection model selection and deployment (Figure 16). The framework organises the decision process across five sequential stages. Stage 1 characterises the dataset properties that determine which strategies are appropriate: fraud prevalence, feature dimensionality and heterogeneity, missingness structure, training data volume, and whether the deployment environment is temporally stationary or subject to concept drift. Stage 2 applies a leakage-safe preprocessing pipeline with joblib caching, ensuring that all statistics are fitted on training data only and that subsequent experimental iterations are computationally efficient. Stage 3 selects the imbalance handling strategy and model family based on the Stage 1 assessment: class weighting with LightGBM as the default for complex

data, with pipeline choice adjusted for cost sensitivity or speed constraints as documented in Table 9. Stage 4 evaluates using the multi-dimensional protocol: PR-AUC and calibration as primary metrics, robustness across seeds, temporal validation if sequential structure exists, and operational timing. Stage 5 translates the evaluation findings into stakeholder-specific deployment recommendations, accounting for regulatory requirements, investigation capacity, and model maintenance constraints.

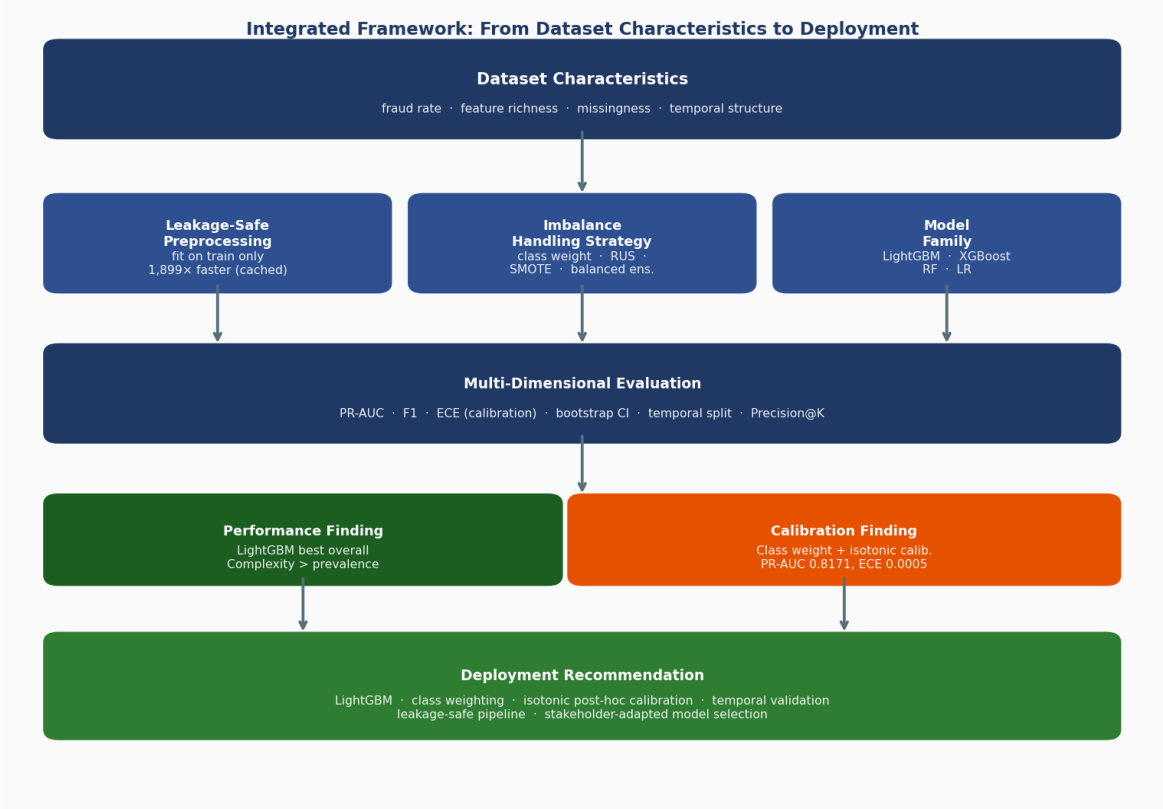


Figure 16: Integrated Framework: From Dataset Characteristics to Deployment Recommendation. The framework synthesises the study's findings across five sequential decision stages, from data characterisation through to stakeholder-specific deployment guidance. Each stage incorporates the empirical findings from Phases 1, 2, and 3.

Table 9 operationalises the framework as a practical decision guide for different organisational contexts, summarising the recommended approach for each stakeholder group based on the empirical evidence in this study.

Table 9: Practical deployment recommendations by organisational context and stakeholder group.

Stakeholder	Model	Imbalance	Calibration	Key Consideration
Banking (regulated)	LightGBM	Class weight	Isotonic calib.	Leakage-safe pipeline; SHAP explainability for audit
E-Commerce / Fintech	LightGBM	Pipeline A	Isotonic calib.	Highest PR-AUC + ECE 0.0005; caching for speed
Cryptocurrency	LightGBM	Class weight	Post-hoc	Forward-chaining temporal validation before deploy
High FN-cost env.	Pipeline D (BalancedRF)	Balanced ens.	—	Lowest missed-fraud count; accept lower PR-AUC
Speed-constrained	LightGBM	RUS (Pipe B)	Platt scaling	50s training; Recall@10% still ~0.85
Research baseline	Logistic Reg.	Class weight	—	Interpretable; min acceptable performance floor

5.3 Conclusion

This study investigated how different machine learning models perform under the class imbalance, data quality, and operational constraints that characterise real-world fraud detection. Across three structurally distinct datasets spanning banking, e-commerce, and cryptocurrency transactions, the findings demonstrate that fraud detection performance is governed less by fraud prevalence than by the interaction between feature richness, imbalance handling strategy, temporal stability, and probability calibration. Phase 2 (3.5% fraud, 627 features) proved substantially more challenging than Phase 1 (0.17% fraud, 30 features) for three of the four evaluated models, directly challenging the common assumption that fraud rate is the primary determinant of detection difficulty.

The results provide a clear answer to the first research question: gradient boosting models consistently outperform classical baselines, with LightGBM offering the strongest overall balance of predictive performance, calibration quality, robustness, and computational efficiency. However, this advantage is conditional on dataset complexity. The difference between LightGBM and XGBoost is negligible on simple, low-dimensional data (0.004 PR-AUC; $p = 0.850$) but becomes substantial on high-dimensional and heterogeneous datasets (0.085 PR-AUC on IEEE-CIS), demonstrating that model selection decisions cannot be separated from the characteristics of the underlying data.

The findings also provide a clear answer to the second research question: feature complexity, missingness patterns, training data volume, and temporal structure exert a greater influence on performance than fraud prevalence alone. The controlled Phase 2 pipeline comparison further demonstrates that class weighting combined with isotonic calibration simultaneously maximises discrimination and probability reliability, while the Phase 3 temporal evaluation quantifies a 0.117 PR-AUC degradation under concept drift that remains invisible under standard random-split protocols. Together, these results show that evaluation design is as consequential as model selection itself.

For practitioners, the principal recommendations are straightforward: adopt LightGBM as the default choice for complex tabular fraud detection problems, use class weighting as the default imbalance strategy, apply post-hoc isotonic calibration when probability-ranked alerting is required, validate temporal generalisation before deployment, and implement leakage-safe preprocessing as standard practice. For researchers, the study highlights the limitations of single-dataset and single-metric benchmarking, demonstrating that reliable conclusions require multi-dataset, multi-pipeline, and operationally grounded evaluation frameworks.

Ultimately, the central contribution of this study is not simply identifying which model performs best but establishing that the conditions under which models are evaluated are as consequential as the models themselves. As fraud patterns continue to evolve and the regulatory scrutiny of automated detection systems intensifies, the ability to select, evaluate, and deploy models with a clear understanding of their data environment will matter as much as algorithmic innovation. This study provides a replicable framework for doing so.

Chapter 6: Limitations and Suggested Future Research

6.1 Limitations

6.1.1 Public Benchmark Datasets and Anonymisation

All three datasets rely on anonymised or PCA-transformed features. This protects privacy and enables public benchmarking but creates two specific constraints. First, feature importance findings cannot be translated into actionable business rules: knowing that V14 is the most discriminative Kaggle feature provides no guidance on which transaction attributes to monitor in production, because the mapping to original features is unknown. Second, the SHAP-based explainability that satisfies regulatory attribution requirements in production systems cannot be demonstrated in practically interpretable form on these datasets.

Practitioners should treat reported PR-AUC values as lower bounds on what is achievable with full feature access and should validate with domain-interpretable features before production deployment.

A related limitation concerns the representativeness of public benchmark fraud rates as proxies for real deployment conditions. The Kaggle dataset's 0.17% fraud rate reflects the prevalence at the time of data collection, but actual fraud rates vary significantly across institutions, card types, and time periods. A model achieving PR-AUC = 0.81 at 0.17% fraud prevalence may perform substantially differently at 0.05% or 0.50% prevalence — a sensitivity documented in the Phase 1 imbalance sensitivity analysis, which shows that all models' PR-AUC estimates shift non-trivially as test-set prevalence is varied between 0.1% and 5%. This means that practitioners who read a PR-AUC value from a benchmark study and apply it directly to their deployment context may be making an implicit assumption of prevalence equivalence that does not hold. Reporting sensitivity analyses alongside primary results, as this study does, provides a more complete picture of the expected performance range across realistic deployment conditions.

6.1.2 No Graph Neural Network or Deep Learning Baselines

This study evaluates four tabular model families but excludes GNN and deep learning architectures. Given the growing literature on GNN-based fraud detection (Xu et al., 2024; Zhang et al., 2025) and the edge list available in the Elliptic dataset capturing the full transaction graph structure, the absence of GNN baselines is a genuine scope limitation. Xu et al. (2024) report PR-AUC improvements of approximately 0.03–0.05 over tabular baselines when applying heterophilic GNNs, suggesting explicit relational modelling may provide additional predictive signal. This comparison is left as a specific direction for future research.

6.1.3 Fixed Hyperparameter Configurations

Experiments use expert-selected fixed configurations rather than formal Bayesian or random search optimisation. This design prevents confounding cross-dataset comparisons with dataset-specific tuning sensitivity, but means reported performance gaps should not be interpreted as maximum achievable differences. The 0.085 Phase 2 LightGBM advantage reflects an architectural difference — leaf-wise versus level-wise growth — rather than a configuration difference and is unlikely to be closed by tuning alone. Logistic Regression is most sensitive to this choice: its conservative regularisation (C=0.1) could plausibly understate its PR-AUC by 0.02–0.05 on complex datasets. Consequently, the study prioritises fair cross-dataset comparison over maximising the performance of individual models, ensuring that observed differences reflect architectural behaviour rather than dataset-specific tuning.

For practitioners seeking to deploy the configurations reported in this study, it is important to understand which hyperparameters are most likely to benefit from dataset-specific tuning. For LightGBM, the most impactful parameters are num_leaves — which controls tree complexity and is set conservatively at 31 for Phases 1 and 3 — and min_child_samples, which constrains the minimum number of examples per leaf and prevents overfitting to rare fraud

patterns. Bayesian optimisation of these two parameters alone is likely to recover a substantial fraction of the performance gap between the fixed and optimally tuned configurations. For XGBoost, max_depth and subsample are the most sensitive parameters under high-dimensional heterogeneous data; the fixed depth of 4 used in this study was selected to prevent memorisation of the small fraud class but may be sub-optimal on the 627-feature IEEE-CIS matrix. For Logistic Regression, the regularisation parameter C has the largest effect: the conservative value of 0.1 suppresses coefficient magnitude aggressively, which improves stability under extreme imbalance but underestimates the model's discriminative capacity on datasets with sufficient fraud examples. Future comparisons that tune these specific parameters while holding others fixed would provide a more complete characterisation of the performance envelope for each model family.

6.1.4 Limited Scope of Resampling Strategies Across All Phases

The controlled pipeline comparison was implemented only for Phase 2, as extending it to all three phases would have tripled the computational requirement without proportional analytical gain. The finding that class weighting outperforms BorderlineSMOTE is therefore specific to the IEEE-CIS context. The Phase 1 regime — 577:1 imbalance, only 344 fraud training cases, and 30 features — represents a setting where the absolute minority-class count is the binding constraint rather than relative prevalence. The literature suggests synthetic oversampling provides greater benefit when the absolute minority count is very small, as interpolation between the few existing examples is more likely to generate in-distribution synthetic samples in a low-dimensional space (Chawla et al., 2002; Al-Daoud and Abu-AlSondos, 2025). Whether BorderlineSMOTE outperforms class weighting in this regime is an open empirical question. Additionally, ADASYN, SMOTE-ENN, and KMeansSMOTE were not evaluated and may behave differently from BorderlineSMOTE on high-dimensional tabular data.

6.1.5 Temporal Evaluation Limited to Phase 3

Forward-chaining temporal evaluation was implemented only for the Elliptic dataset, which provides unambiguous discrete time-step labels. The IEEE-CIS TransactionDT and Kaggle Time features both reference undisclosed origin times, making unbiased temporal splits infeasible without assumptions that could introduce systematic evaluation bias. Phase 1 and Phase 2 PR-AUC scores are therefore random-split estimates that may overstate deployment performance if concept drift is present. Given that the e-commerce fraud context is generally considered more temporally dynamic than card fraud, temporal robustness evaluation is arguably more important for Phase 2 than Phase 3 — a gap that future work implementing approximate temporal splits on the IEEE-CIS dataset could partially address.

6.1.6 Absence of Real-Time Streaming Evaluation

This study evaluates models under offline batch conditions using static, fully labelled datasets. Real-world fraud detection additionally faces streaming latency constraints (50–200ms budgets per transaction), delayed label arrival (fraud confirmation can take days or weeks, creating periods where the model cannot receive feedback), and feedback loops where the deployed model shifts the fraud distribution it is subsequently trained on. These deployment realism gaps are common to all benchmark-based research but bound the generalisability of results to controlled evaluation conditions rather than live production systems.

6.2 Suggested Future Research

Eight directions emerge from the limitations and findings of this study.

First, a systematic comparison of ADASYN, SMOTE-ENN, KMeansSMOTE, and cost-sensitive learning across all three datasets under the same controlled N=10 protocol would clarify when synthetic oversampling benefits discrimination versus harms calibration,

extending the Phase 2 pipeline evidence to Phase 1 and Phase 3 regimes where the present study cannot generalise.

Second, temporal validation should be expanded to all three datasets using rolling-window and forward-chaining cross-validation protocols, providing more realistic deployment performance estimates and revealing how quickly each model family degrades under concept drift of different speeds and magnitudes.

Third, a direct comparison against GNN-based approaches using the Elliptic edge list — under the same leakage-safe protocol and train-test splits — would quantify the added value of explicit relational modelling under controlled conditions, directly addressing the theory-practice gap documented in Section 2.1.

Fourth, the isotonic calibration finding (ECE = 0.0005 while preserving PR-AUC) should be tested for generalisability across additional fraud datasets and imbalance conditions, including whether Platt scaling or temperature scaling offer comparable calibration benefits at lower computational cost.

Fifth, future research should incorporate real operational cost functions derived from actual investigation capacity, chargeback losses, and customer friction costs, making model selection guidance more directly actionable for financial institutions.

Sixth, future research should investigate continual and adaptive learning approaches designed to maintain fraud detection performance under evolving fraud patterns. The temporal evaluation in Phase 3 demonstrated a 0.117 PR-AUC degradation under concept drift, confirming that static models become progressively less reliable when deployed on future data. Continual learning frameworks that update model parameters while preserving previously acquired knowledge may provide a practical compromise between full retraining and static deployment. Comparing adaptive boosting approaches, incremental tree-based

learning, and continual learning strategies would help determine whether performance degradation can be mitigated while controlling computational and operational costs.

Seventh, online and active learning frameworks represent the most direct response to the concept drift challenge quantified in Phase 3. Comparing periodic batch retraining at different intervals against continuous online gradient updates and evaluating active labelling strategies under a fixed investigation budget per time step, would provide concrete guidance on when and how to retrain — the most operationally consequential question practitioners face in production.

Eighth, the integration of fairness constraints deserves investigation. Fraud detection models may systematically assign different false-positive rates across demographic groups, raising both ethical and regulatory concerns. Future work using richer proprietary datasets should assess whether the optimal configurations identified in this study maintain fairness across subgroups, and whether the imbalance handling strategies that optimise PR-AUC do so uniformly or by concentrating gains in the majority subgroup.

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Appendices

Appendix A – Experimental Configuration Summary

This study evaluates four machine learning models across three fraud detection datasets:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

Evaluation Framework

Component	Description
Baseline benchmarking	PR-AUC, Average Precision, F1 Score, Precision, Recall, ECE, Brier Score, Training Time
Robustness analysis	Repeated experiments across multiple random seeds
Dataset-size sensitivity	Training fractions : 1%, 5%, 10%, 25%, 50%, 100%
Imbalance sensitivity	Multiple fraud prevalence levels
Calibration analysis	Reliability diagrams, ECE, Brier Score, post-hoc isotonic calibration
Statistical significance testing	Paired t-tests, Wilcoxon signed-rank tests, Bonferroni correction
Explainability	SHAP feature importance and summary plots

Experimental Repetitions

Phase	Repeated Runs
Phase 1	N = 5
Phase 2 (original framework)	N = 5

Phase 2 (improved framework)	N = 10
Phase 3	N = 5

Class Imbalance Handling

Phase	**Strategy**
Phase 1	Class weighting
Phase 2 (original framework)	Random UnderSampling + BorderlineSMOTE
Phase 2 – Pipeline A	Class weighting only
Phase 2 – Pipeline B	Random UnderSampling
Phase 2 – Pipeline C	Random UnderSampling + BorderlineSMOTE
Phase 2 – Pipeline D	Balanced ensemble methods
Phase 3	Class weighting

Appendix B – Dataset Summary

Dataset	Domain	Rows	Features	Fraud Rate
Kaggle Credit Card Fraud	Banking	283,726	30	0.1727%
IEEE-CIS Fraud Detection	E-commerce	590,540	627	3.4990%
Elliptic Bitcoin Transactions	Cryptocurrency	203,769[b]	165	9.76%

[b] Elliptic: 203,769 total transactions; 46,564 carry ground-truth labels and are used in all analyses only. The remaining 157,205 unlabelled transactions are excluded to avoid selection bias.

Appendix C – Statistical Significance Testing

Phase 1 – Paired t-tests (N = 5)

Bonferroni-corrected significance level:

$\alpha = 0.0083$

Comparison	p-value	Significant after Bonferroni?
Logistic Regression vs Random Forest	0.0143	No
Logistic Regression vs XGBoost	0.0118	No
Logistic Regression vs LightGBM	0.0064	Yes
Random Forest vs XGBoost	0.0429	No
Random Forest vs LightGBM	0.0104	No
XGBoost vs LightGBM	0.8499	No

Only the comparison between Logistic Regression and LightGBM remained significant after Bonferroni correction.

Phase 2 – Improved Framework (N = 10)

Wilcoxon Signed-Rank Test

Bonferroni-corrected significance level:

$\alpha = 0.0167$

Comparison	p-value	Significant after Bonferroni ?
Pipeline A vs Pipeline B	0.0020	Yes
Pipeline A vs Pipeline C	0.0020	Yes
Pipeline B vs Pipeline C	0.0020	Yes

All pipeline comparisons remained statistically significant after correction.

Phase 3 – Paired t-tests (N = 5)

Bonferroni-corrected significance level:

$\alpha = 0.0083$

Comparison	**p-value**	**Significant after Bonferroni ?**
Logistic Regression vs Random Forest	< 0.001	Yes
Logistic Regression vs XGBoost	< 0.001	Yes
Logistic Regression vs LightGBM	< 0.001	Yes
Random Forest vs XGBoost	0.0001	Yes
Random Forest vs LightGBM	< 0.001	Yes
XGBoost vs LightGBM	0.001	Yes

All six pairwise comparisons remained significant after correction.

Appendix D – Supplementary Figures

This appendix contains figures generated during the analysis that provide supporting evidence for the results discussed in Chapter 4 but were not included in the main body of the dissertation to preserve narrative focus.

- Figure D1 – Phase 1 Confusion Matrices

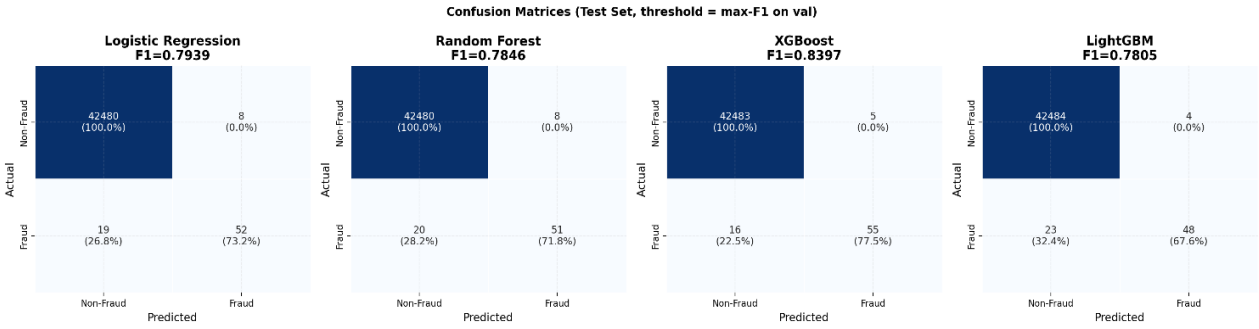


Figure D1. Confusion matrices for the four evaluated models in Phase 1 using the F1-optimised decision threshold. The matrices illustrate the trade-off between fraud detection performance and false-positive generation.

- Figure D2 – Phase 1 Dataset-Size Sensitivity Analysis

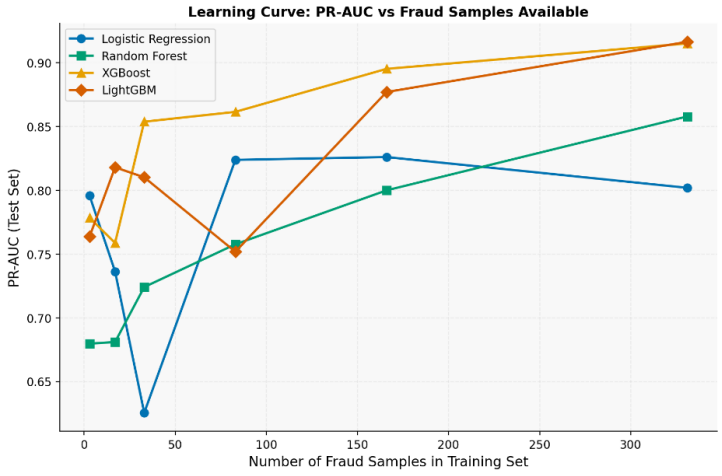


Figure D2. Dataset-size sensitivity analysis showing PR-AUC as a function of the number of fraud samples available during training. Gradient boosting models benefit substantially from additional training data, while Logistic Regression exhibits comparatively limited gains.

- Figure D3 – Phase 1 Reliability Diagrams

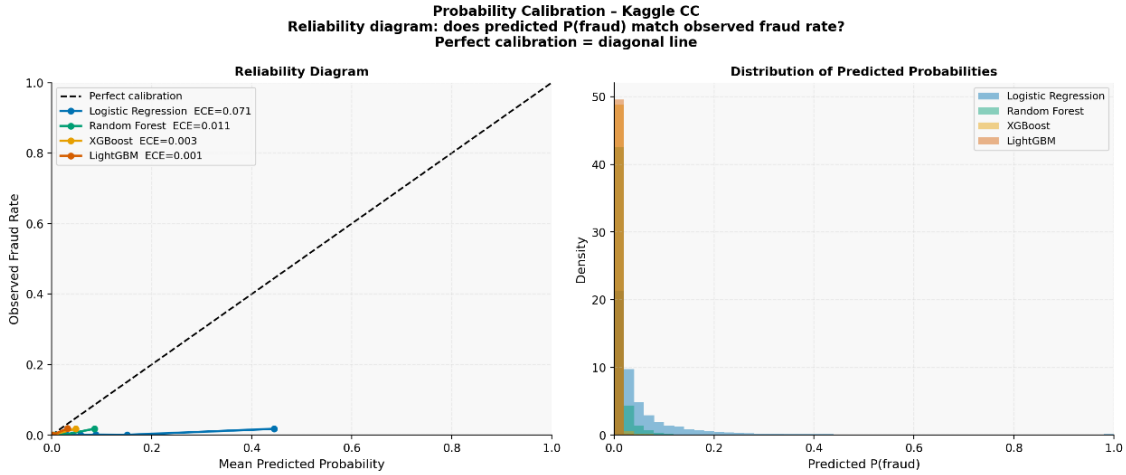


Figure D3. Reliability diagrams and predicted probability distributions for Phase 1. LightGBM (ECE = 0.0014) and XGBoost (ECE = 0.0029) exhibit the strongest calibration performance, while Logistic Regression shows the largest calibration deviation (ECE = 0.0715).

- Figure D4 – Phase 2 Missing Values Analysis

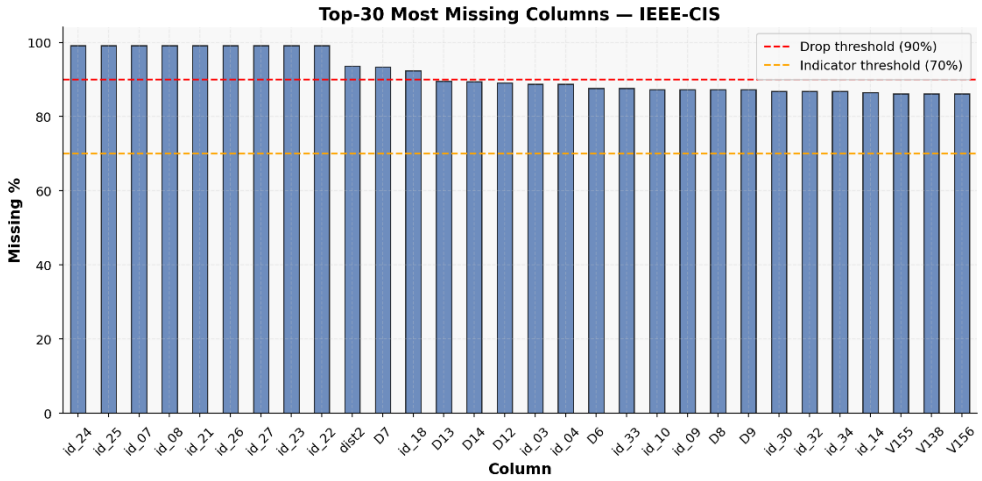


Figure D4. Percentage of missing values in the most incomplete IEEE-CIS features. Several variables exceed 90% missingness, motivating the leakage-safe missing-value handling strategy adopted in Phase 2.

- Figure D5 – Phase 2 Precision@K and Recall@K Curves

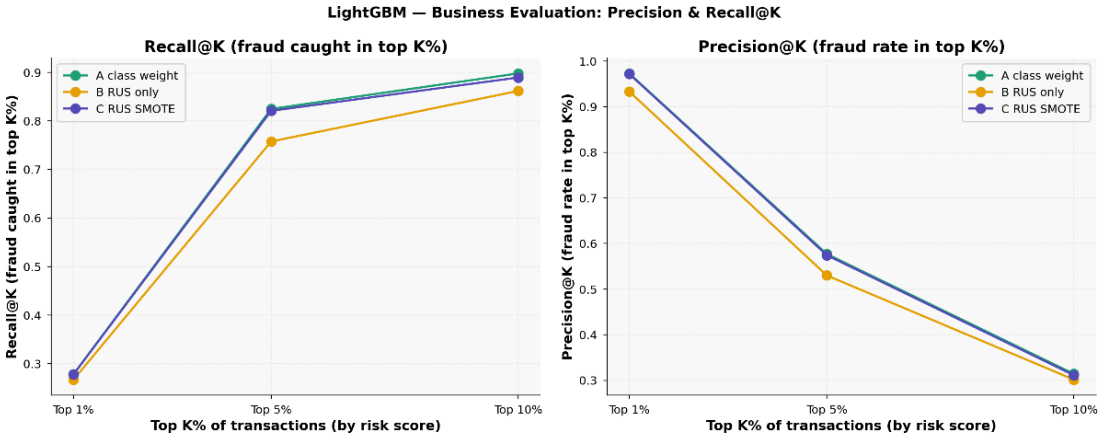


Figure D5. Business-oriented evaluation using Precision@K and Recall@K metrics. Pipeline A (class weighting) consistently achieves the strongest recall and precision among the evaluated imbalance-handling strategies.

- Figure D6 – Phase 2 Cost-Sensitive Evaluation

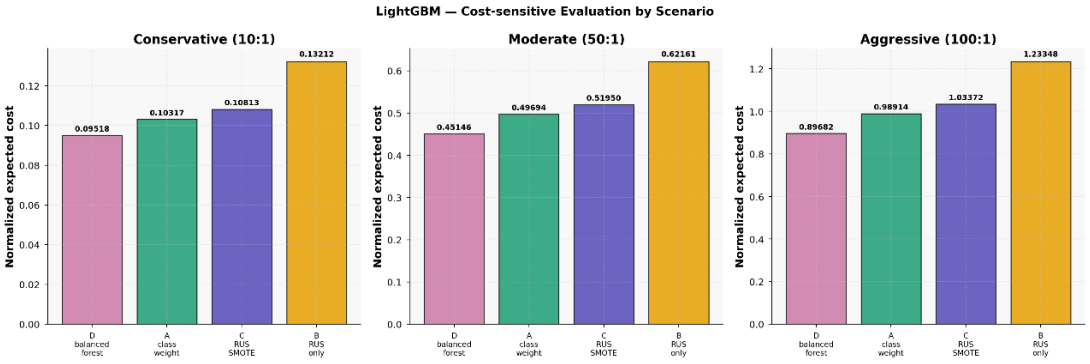


Figure D6. Cost-sensitive evaluation of the four Phase 2 pipelines under different false-negative to false-positive cost ratios. Pipeline A remains the most effective configuration across all operational scenarios.

- Figure D7 – Phase 2 SHAP Feature Importance Ranking

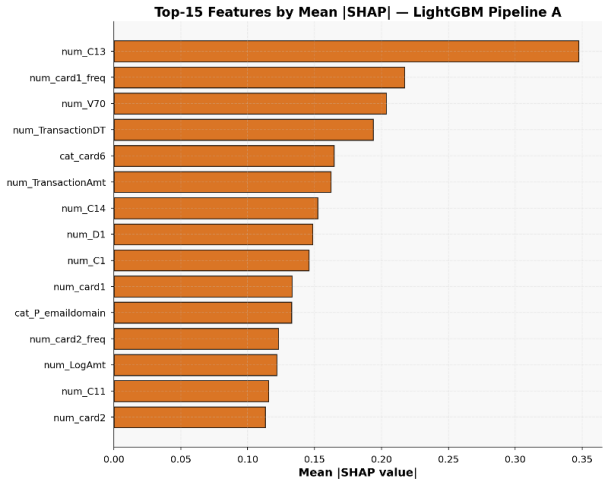


Figure D7. Mean absolute SHAP values for the most influential features in the Phase 2 LightGBM model. Transaction frequency and temporal variables dominate the importance ranking, indicating their central role in fraud prediction.

- Figure D8 – Additional Robustness Analysis Results

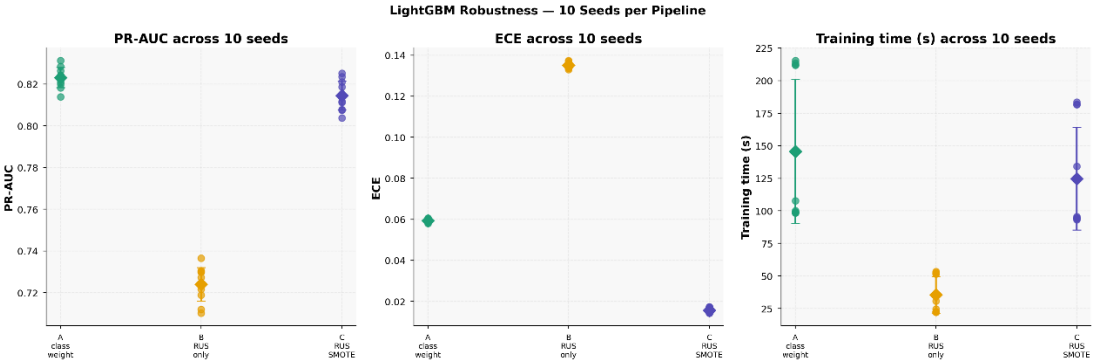


Figure D8. Robustness analysis of the improved Phase 2 framework across ten independent experimental runs. The distributions of PR-AUC, calibration error (ECE), and training time demonstrate the stability and reproducibility of the proposed approach.

- Figure D9 – Additional Feature Importance Visualisations

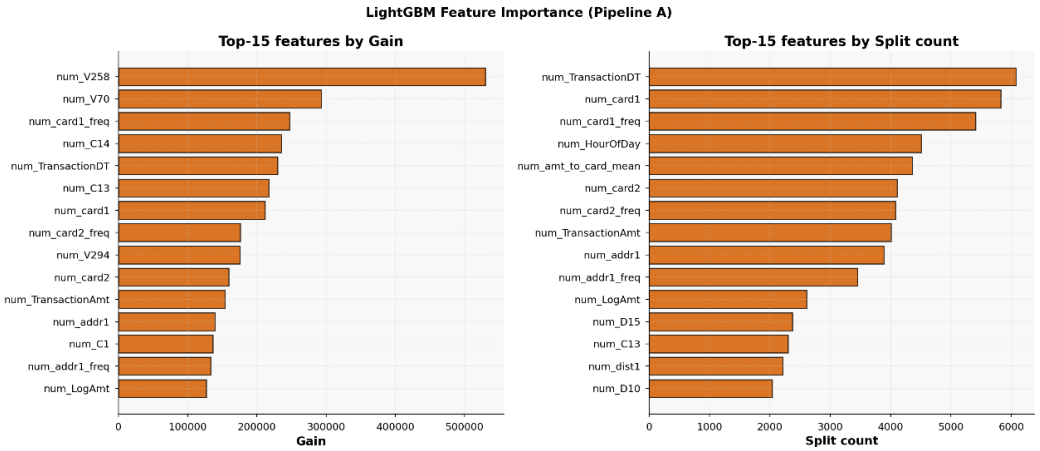


Figure D9. LightGBM feature importance rankings based on gain and split count. The figure provides a complementary interpretability perspective to the SHAP analysis presented in Figure D7 and confirms the importance of transaction frequency and temporal variables.

All figures can be reproduced using the accompanying Jupyter notebooks and the random seed configuration specified in Appendix E.

Appendix E – Reproducibility and Scalability Information

Software Environment

Component	Version
Python	3.10
NumPy	1.26.4
Pandas	2.0
scikit-learn	1.3
XGBoost	2.0
LightGBM	4.6.0
imbalanced-learn	0.11
SHAP	0.46.0
matplotlib	3.7
seaborn	0.12
joblib	1.3

Random Seed Configuration

Primary experiments:

- Seed = 42

Robustness experiments:

- Seeds 42–51

Random seeds were fixed for all experiments to ensure reproducibility.

Scalability Benchmark Results

The scalability improvements introduced in Phase 2 constitute an important practical contribution to this research.

Data Loading Optimisation

Method	**Loading Time**
CSV parsing and table joins	31.3 seconds
Pickle deserialisation	2.12 seconds

Result: approximately **15× faster** loading.

Preprocessing Optimisation

Method	**Preprocessing Time**
Full preprocessing pipeline	170.9 seconds
Cached preprocessing retrieval	0.09 seconds

Result: 1,899× faster preprocessing.

By combining pickle-based dataset caching, float32 memory optimisation, and joblib preprocessing caching, the study substantially reduced computational overhead while preserving leakage-safe preprocessing and full experimental reproducibility. These optimisations enabled extensive robustness experiments involving multiple pipelines, models, and random seeds that would otherwise have been impractical to execute within reasonable time constraints.